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Abstract

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1 The Hilbert–Schmidt paradox beyond Gaussian symmetry

The Feldman–Hájek theorem assigns a remarkably rigid boundary to a Gaussian perturbation. After the Cameron–Martin spaces have been identified, equivalence of two Gaussian laws is decided by a Hilbert–Schmidt condition on their relative covariance; mean displacement is decided by the Cameron–Martin norm [3, 4, 1]. In a whitened chart this takes the familiar form

$$P \in \mathcal{S}_2(H), \qquad h \in H.$$

At first sight, the class $\mathcal{S}_2$ appears to be a specifically Gaussian accident. For a non-Gaussian product law there is no covariance operator from which the same ideal can simply be read off, and an arbitrary linear change of coordinates destroys the product decomposition on which Kakutani's theorem acts.

The wrong response is to guess a replacement ideal. The more invariant question is instead:

which topology on a transformation orbit keeps measure class stable under completion?

The operator ideal, when one exists, should be the expression of that topology in a particular local chart. From this point of view the Gaussian $\mathcal{S}_2$ condition is an answer in coordinates, not the definition of the problem.

1.1 The symmetry hidden by covariance notation

Let a group G act measurably on a standard Borel space X, and fix a probability measure μ. Two subgroups have different roles:

$$\mathrm{Stab}(\mu) := \{g \in G : g_\# \mu = \mu\}, \qquad \mathrm{QStab}(\mu) := \{g \in G : g_\# \mu \sim \mu\}.$$

The orbit map $g \mapsto g_\# \mu$ is constant on the right cosets of $\operatorname{Stab}(\mu)$, since

$$(gh)_\# \mu = g_\#(h_\# \mu) = g_\# \mu \quad \text{for } h \in \operatorname{Stab}(\mu).$$

Thus the law orbit is naturally parametrized by $G/\operatorname{Stab}(\mu)$, whereas the finite-distance branch sought in a Feldman–Hájek-type theorem is $\operatorname{QStab}(\mu)/\operatorname{Stab}(\mu)$. Confusing G with this quotient creates spurious directions and leads to false necessity statements.

For the standard Gaussian law in $\mathbb{R}^d$, the linear stabilizer is the full orthogonal group. Its Lie algebra consists of all skew-symmetric matrices, so infinitesimal rotations are invisible to likelihood. The observable covariance direction is the quotient

$$\mathfrak{gl}_d/\mathfrak{o}_d \simeq \operatorname{Sym}_d.$$

In fixed dimension this quotient is already complete. On a separable Hilbert space, however, the directed union over finite coordinate subspaces completes in the Gaussian Fisher norm to the self-adjoint Hilbert–Schmidt class. This is the symmetry mechanism behind the classical coordinate $\mathcal{S}_2$.

For a generic non-Gaussian i.i.d. law, it is inaccurate to say that the stabilizer disappears. Coordinate permutations remain exact symmetries, and an even one-dimensional density also permits sign changes. What typically disappears is the connected, or Lie, part of the stabilizer. The remaining symmetry group is often discrete; hence its Lie algebra is zero. Infinitesimal rotations that were null directions in the Gaussian model then acquire positive Fisher cost. The finite-dimensional Fisher operator computed below makes this collapse quantitative: its skew-symmetric eigenvalue is zero at the Gaussian density and positive away from it under the stated regularity assumptions.

This symmetry calculation explains a *local* change of geometry. It does not by itself establish a global quasi-invariance theorem. In particular, even a coercive Fisher form at the identity does not imply

$$T_\# \mu \sim \mu \quad \Longleftrightarrow \quad T - I \in \mathcal{S}_2.$$

The displayed equivalence is already false without taking the exact stabilizer into account: an infinite coordinate permutation can preserve an i.i.d. law although its difference from the identity is not Hilbert–Schmidt. The invariant candidate, if valid for a specified model, must have the form

$$T_\# \mu \sim \mu \quad \Longleftrightarrow \quad \text{the coset } T \operatorname{Stab}(\mu) \text{ has finite information length.}$$

Only after that length has been computed in a uniformly regular chart may it reduce to a condition such as

$$\text{there exists } Q \in \operatorname{Stab}(\mu) \text{ such that } TQ - I \in \mathcal{S}_2.$$

Even then this is a description of one finite-distance component, not a globally finite metric on all of G.

1.2 Why the endpoint problem is harder than the tangent problem

Four logically separate issues intervene between a Fisher calculation and an endpoint equivalence statement.

(i) An infinite matrix must define a measurable transformation on the space carrying the product law, and its inverse must be meaningful on the same measurable model.

(ii) General linear mixing destroys independent blocks. Consequently the exact product dichotomy is no longer available merely because the reference coordinates were initially independent.

(iii) A quadratic expansion of finite-dimensional Hellinger affinity must be uniform as the projection rank grows. Pointwise differentiability in each dimension does not prevent information from escaping through the tail.

(iv) Projected likelihood ratios form martingales, but convergence of their finite-dimensional laws does not by itself preserve a global Radon–Nikodým derivative. One needs an equivalence–singularity dichotomy or two-sided uniform integrability to close the likelihood and reciprocal-likelihood graphs.

These are global completion questions. For the regular even i.i.d. density class specified in equation (3.1), this paper resolves all four and proves the arbitrary-mixing criterion theorem 3.1. The density assumptions are explicit; the result does not assert the same criterion for every non-Gaussian law.

1.3 Organization and scope

The argument proceeds in five stages. The present section formulates the apparent conflict between the universal Gaussian threshold and the lawdependent non-Gaussian geometry. The next section reconstructs the Gaussian theorem backwards: first as an affine orbit modulo its orthogonal stabilizer, then as a Fisher completion, and finally as the classical Feldman–Hájek endpoint statement.

The third section studies a fixed i.i.d. density. After the exact block criteria and the full finite-matrix Fisher spectrum, it proves a dimension-free nonlinear Hellinger comparison. Tail cumulant rigidity, a signed-permutation matching argument, and two-sided likelihood closure then give the arbitrary-mixing endpoint theorem under equation (3.1).

The fourth section removes the statistical coordinates from the argument. It builds a projective uniformity from finite tests and separates it from a complete likelihood-graph metric on each reference measure class. The finite tests identify points, the Hellinger-energy net diagnoses the global class relation, and the likelihood graph supplies the class-preserving completion. The fifth section returns to geometry and identifies the kernel of the Fisher form with the Lie algebra of the exact stabilizer. Gaussian covariance geometry then appears as the maximally symmetric case, not as the template from which every non-Gaussian model must inherit an operator ideal.

2 Gaussian Feldman–Hájek theory reconstructed from the quotient

The usual statement of the Gaussian theorem begins with two means and two covariances. For the later non-Gaussian comparison it is more revealing to reverse that order. We first identify a Gaussian law as a point of an affine transformation orbit, quotient out transformations that leave the law fixed, and differentiate the resulting orbit. Covariance and the Hilbert–Schmidt condition then reappear as coordinates on this quotient.

2.1 The finite-dimensional affine orbit is a right quotient

Let

$$G_d = \mathrm{GL}_d(\mathbb{R}) \ltimes \mathbb{R}^d, \qquad (T,b)\cdot x = Tx + b,$$

with multiplication

$$(T,b)(S,c) = (TS, Tc + b).$$

Write $\gamma_d = \mathcal{N}(0, I_d)$, and embed $O(d)$ in G_d as

$$H_d := \{(Q,0) : Q \in O(d)\}.$$

Proposition 2.1 (Gaussian laws as an affine quotient). *The stabilizer of γ_d in G_d is H_d, and the orbit map induces a bijection*

$$\begin{aligned} G_d/H_d &\longrightarrow \{\textit{nondegenerate Gaussian laws on } \mathbb{R}^d\}, \\ (T,b)H_d &\longmapsto \mathcal{N}(b, TT^*). \end{aligned}$$

Equivalently,

$$G_d/H_d \simeq \mathrm{Sym}_d^{++} \times \mathbb{R}^d, \qquad (T,b)H_d \longmapsto (TT^*, b).$$

Here the quotient is a right quotient.

Proof. The pushforward of γ_d by (T,b) is $\mathcal{N}(b, TT^*)$. It equals γ_d exactly when $b = 0$ and $TT^* = I_d$, which is equivalent to $T \in O(d)$. If two affine maps give the same Gaussian law, then their translations agree and $TT^* = SS^*$. Consequently $Q = T^{-1}S$ is orthogonal and $(S,b) = (T,b)(Q,0)$. Conversely, right multiplication by $(Q,0)$ does not change the pushforward. Finally, polar decomposition in the form $T = (TT^*)^{1/2}Q$ selects the unique positive-definite representative of each linear coset. □

The right-coset convention is structural rather than cosmetic. An element of the stabilizer acts first:

$$((T,b)(Q,0))_{\#}\gamma_d = (T,b)_{\#}((Q,0)_{\#}\gamma_d) = (T,b)_{\#}\gamma_d.$$

Thus a condition placed directly on $T - I$, without minimizing or quotienting over the right orthogonal action, cannot be an invariant condition on Gaussian laws.

2.2 Differentiating the quotient

Consider a differentiable curve through the identity,

$$T_t = I_d + tA + o(t), \qquad b_t = tv + o(t),$$

where $A \in \mathfrak{gl}_d$ and $v \in \mathbb{R}^d$. Let p_t be the density of $(T_t, b_t)_{\#}\gamma_d$ with respect to Lebesgue measure. Differentiating

$$p_t(x) = |\det T_t|^{-1}(2\pi)^{-d/2} \exp\left[-\frac{1}{2}\left|T_t^{-1}(x - b_t)\right|^2\right]$$

at $t = 0$ gives the score

$$\dot{\ell}_{A,v}(x) := \frac{d}{dt}\bigg|_{t=0} \log p_t(x) = x^* A x - \operatorname{tr} A + \langle v, x \rangle.$$

Writing

$$S = \frac{A + A^*}{2}, \qquad K = \frac{A - A^*}{2},$$

we obtain

$$\dot{\ell}_{A,v}(x) = \langle S, xx^* - I_d \rangle_F + \langle v, x \rangle.$$

The skew part K has vanished before any norm is taken.

Proposition 2.2 (Gaussian Fisher spectrum)**.** *For $X \sim \gamma_d$,*

$$\mathcal{I}_{\gamma_d}(A, v) := \mathbb{E}\big[\dot{\ell}_{A,v}(X)^2\big] = 2\|S\|_F^2 + \|v\|^2.$$

Relative to the orthogonal decomposition

$$\mathfrak{gl}_d \ltimes \mathbb{R}^d = \mathrm{Sym}_d \oplus \mathfrak{o}_d \oplus \mathbb{R}^d,$$

the Fisher operator has eigenvalue 2 *on* Sym_d*, eigenvalue* 0 *on* $\mathfrak{o}_d$*, and eigenvalue* 1 *on the translation sector. In particular,*

$$\ker \mathcal{I}_{\gamma_d} = \mathfrak{o}_d = \mathrm{Lie}(H_d).$$

Hence Fisher information is nondegenerate precisely on

$$(\mathfrak{gl}_d \ltimes \mathbb{R}^d)/\mathfrak{o}_d \simeq \mathrm{Sym}_d \oplus \mathbb{R}^d.$$

Proof. For a standard Gaussian vector, Wick's formula yields

$$\mathbb{E}\big[\langle S, XX^* - I_d \rangle_F^2\big] = 2\|S\|_F^2.$$

The centered quadratic term is orthogonal to the linear term, and $\mathbb{E}\langle v, X \rangle^2 = \|v\|^2$. This proves the formula and the three spectral assertions. The equality of the kernel with $\mathrm{Lie}(H_d)$ is also the differential of the exact orthogonal invariance established above. □

This calculation already contains the reason that Gaussian covariance perturbations see self-adjoint Hilbert–Schmidt operators rather than all Hilbert–Schmidt operators. The missing skew sector is not small; it is an entire stabilizer direction that has been quotiented out.

2.3 The isonormal tangent completion

Let H be an infinite-dimensional separable real Hilbert space and let

$$W : H \longrightarrow L^2(\Omega, \mathcal{G}, \mathbb{P})$$

be an isonormal Gaussian process: $W(h)$ is centered and $\mathbb{E}[W(h)W(k)] = \langle h, k\rangle_H$. This formulation is essential. There is no H-valued Gaussian random variable with covariance I_H when H is infinite-dimensional, because I_H is not trace class. Accordingly, the expressions below concern the joint laws of the Gaussian coordinates $W(h)$; they do not assert the existence of an H-valued $\mathcal{N}(0, I_H)$.

For a finite-rank operator A and $v \in H$, set $S = (A + A^*)/2$. The finite-dimensional calculation on any subspace containing the ranges of A, A^*, and the vector v gives the isonormal score

$$\dot{\ell}_{A,v} = I_2(S) + W(v),$$

where $I_2(S) = I_2(S)$ is the Wick-centered quadratic form; here I_2 denotes the second Wiener integral over the isonormal process W, i.e. the element of its second Wiener chaos with the stated spectral form. If $Se_j = s_j e_j$, then

$$I_2(S) = \sum_j s_j \big(W(e_j)^2 - 1\big)$$

for finite-rank S, and by L^2-closure for self-adjoint Hilbert–Schmidt S. The first and second homogeneous Wiener chaoses are orthogonal, and the standard chaos isometry gives [5]

$$\|\dot{\ell}_{A,v}\|^2_{L^2(\mathbb{P})} = 2\|S\|^2_{\mathcal{S}_2} + \|v\|^2_H.$$

Let $\mathfrak{gl}_{\mathrm{fin}}(H)$ denote the finite-rank operators and let $\mathfrak{o}_{\mathrm{fin}}(H)$ denote their skew-adjoint subspace. The preceding formula defines a norm on

$$(\mathfrak{gl}_{\mathrm{fin}}(H)/\mathfrak{o}_{\mathrm{fin}}(H)) \oplus H.$$

Its Hilbert completion is canonically

$$\boxed{\mathcal{S}_{2,\mathrm{sa}}(H) \oplus H, \qquad \|(S, v)\|^2_{\mathrm{Fisher}} = 2\|S\|^2_{\mathcal{S}_2} + \|v\|^2_H.}$$

Thus the Gaussian $\mathcal{S}_2$ chart is obtained by the sequence

finite affine transformations $\longrightarrow$ quotient by orthogonal symmetry $\longrightarrow$ Fisher completion.

It is not obtained by declaring an operator ideal before the likelihood has identified its null directions.

At the level of Gaussian coordinate laws, every $Q \in O(H)$ preserves the isonormal law because $\langle Qh, Qk \rangle = \langle h, k \rangle$. One need not assume that each such coordinate symmetry is represented by a point transformation of the particular probability space $(\Omega, \mathcal{G}, \mathbb{P})$. The quotient statement is a statement about laws, which is the level relevant to Feldman–Hájek equivalence.

2.4 The global endpoint theorem in relative coordinates

We now reconnect the quotient calculation with Gaussian measures on an ambient Hilbert space. Let E be a separable real Hilbert space and

$$\mu_i = \mathcal{N}(m_i, C_i), \qquad i = 0, 1,$$

where C_i are positive injective trace-class operators. Their Cameron–Martin spaces, viewed as subsets of E, are

$$H_i = \operatorname{Ran}(C_i^{1/2}).$$

The inverse $C_0^{-1/2}$ is initially an unbounded operator with domain $\operatorname{Ran}(C_0^{1/2})$. If $H_0 = H_1$, the composition

$$L = C_0^{-1/2} C_1^{1/2}$$

is everywhere defined, and the closed graph theorem, applied in both directions, gives a bounded invertible operator on the whitened Hilbert space. Equivalently,

$$C_1 = C_0^{1/2} L L^* C_0^{1/2}$$

as a covariance factorization. The positive operator

$$R_{10} := LL^*$$

is the relative covariance. More precisely, if the canonical factors are replaced on the right by orthogonal operators U_0, U_1, then the transition operator becomes $L' = U_0^* L U_1$ and $L' L'^* = U_0^* R_{10} U_0$. Thus the numerator factor cancels, the denominator factor conjugates, and the condition $R_{10} - I \in \mathcal{S}_2$ is invariant.

Theorem 2.3 (Feldman–Hájek in quotient coordinates)**.** *The measures μ_0 and μ_1 are either equivalent or mutually singular. They are equivalent if and only if* *(i) their Cameron–Martin ranges agree:* $H_0 = H_1$*;*

(ii) their mean difference belongs to the common range: $m_1 - m_0 \in H_0$*;*

(iii) their relative covariance satisfies $R_{10} - I \in \mathcal{S}_2$*.*

If any one of these conditions fails, the measures are mutually singular.

This is the classical Feldman–Hájek theorem [3, 4, 1]. The first condition places the two covariance factors in the same invertible transformation orbit. The third measures the positive part of that orbit after the irrelevant right-orthogonal factor has been removed. The second is the Cameron–Martin translation condition [2, 1].

There is a useful factor version of the same statement. If $C_i = B_i B_i^*$ are represented by injective covariance factors $B_i : K \to E$ on a common source Hilbert space and their ranges agree, one may write $B_1 = B_0 T$ with T bounded and invertible on the source Hilbert space. Then the relative covariance is TT^*, up to orthogonal conjugacy, and the covariance criterion is

$$TT^* - I \in \mathcal{S}_2.$$

The expression depends only on the right coset $TO(K)$. By polar decomposition that coset has positive representative $(TT^*)^{1/2}$, and

$$TT^* - I \in \mathcal{S}_2 \quad \iff \quad (TT^*)^{1/2} - I \in \mathcal{S}_2,$$

because the positive representative is bounded and boundedly invertible. This is the global counterpart of the quotient tangent space $\mathcal{S}_{2,\mathrm{sa}}$.

2.5 Recovery of the P-chart

The covariance chart used in the Gaussian likelihood analysis is recovered by fixing a positive injective trace-class covariance C, choosing a bounded self-adjoint P, and assuming

$$I - P \geq \rho I \qquad \text{for some } \rho > 0.$$

Set

$$T_t = (I - tP)^{1/2}, \qquad C_t = C^{1/2} T_t T_t^* C^{1/2} = C^{1/2}(I - tP)C^{1/2}, \qquad 0 \leq t \leq 1.$$

The lower bound makes every T_t boundedly invertible, so all C_t have the same Cameron–Martin range. The right-quotient relative covariance at the endpoint is exactly

$$T_1 T_1^* = I - P.$$

Consequently Feldman–Hájek reduces to

$$\mathcal{N}(m, C) \sim \mathcal{N}\Big(m + \delta m, C^{1/2}(I - P)C^{1/2}\Big) \iff P \in \mathcal{S}_2, \quad \delta m \in \mathrm{Ran}(C^{1/2}).$$

Differentiating the factor path at zero gives

$$A = \dot{T}_0 = -\frac{1}{2}P.$$

To avoid presupposing that P is Hilbert–Schmidt, let (Π_n) be increasing finite-rank orthogonal projections converging strongly to the identity on the source space E, and put $P_n = \Pi_n P \Pi_n$. Taking $H = E$ in the preceding isonormal construction, the finite-shell scores are

$$S_{0,n} = -\frac{1}{2}I_2(P_n).$$

Wick's isometry gives

$$\|S_{0,n} - S_{0,m}\|^2_{L^2(\mathbb{P})} = \frac{1}{2}\|P_n - P_m\|^2_{\mathcal{S}_2}.$$

Hence $(S_{0,n})$ is L^2-Cauchy if and only if $P \in \mathcal{S}_2$: the forward implication follows from finite-rank approximation in $\mathcal{S}_2$, while in the reverse direction the Hilbert–Schmidt limit of (P_n) must equal its strong-operator limit P. In that case the score limit

$$S_0 = -\frac{1}{2}I_2(P) \quad \text{satisfies} \quad \|S_0\|^2_{L^2(\mathbb{P})} = \frac{1}{2}\|P\|^2_{\mathcal{S}_2}.$$

If $\delta m = C^{1/2}h$, then $h \in E$; the translation score is $W(h)$ for the source isonormal process and has Fisher norm $\|h\|_E$. We have thus recovered, from the quotient description,

$$(S_{0,n}) \text{ is Cauchy in } L^2 \iff P \in \mathcal{S}_2 \iff \text{the centered endpoint laws are equivalent,}$$

with the Cameron–Martin condition added for means.

The coincidence of the local and endpoint thresholds is exceptionally clean in the Gaussian model. Orthogonal symmetry removes the entire skew sector, Wick's isometry turns the remaining quadratic score into the self-adjoint Hilbert–Schmidt norm, and the global Gaussian dichotomy closes the endpoint. The next sections retain this logical order while separating the parts that survive for a non-Gaussian reference law from the parts that depend on Gaussian symmetry and exact quadratic structure.

3 An i.i.d. non-Gaussian equivalence criterion

For a regular non-Gaussian i.i.d. law, arbitrary bounded invertible linear mixing preserves measure class exactly when it is a Hilbert–Schmidt perturbation of a signed permutation. The proof uses nonlinear Hellinger estimates, rigidity of the tail cumulants, and a measurable Radon–Nikodym closure. Independence of the transformed coordinates is never assumed.

3.1 The density class and the endpoint theorem

The arbitrary-mixing result below uses the following explicit assumptions:

$$\begin{gathered} f(x) = Z^{-1}e^{-V(x)}, \qquad V \in C^2(\mathbb{R}), \qquad V(-x) = V(x), \qquad \|V''\|_\infty \le L < \infty, \\ \mathbb{E}_f X^2 = 1, \qquad \mathbb{E}_f|X|^p < \infty \quad (1 \le p < \infty), \qquad f \ne (2\pi)^{-1/2}e^{-x^2/2}. \end{gathered} \tag{3.1}$$

These are conditions on the one-dimensional density, not assumptions of Hellinger coercivity, tail rigidity, or likelihood closure. For example, the densities proportional to $\exp(-x^2/2 - a\cos x)$, $0 < |a| < 1$, after rescaling to variance one, satisfy them. The all-moments assumption can be weakened to a finite moment condition stated in section 3.8 below.

Put $H = \ell^2(\mathbb{N};\mathbb{R})$ and $\mu = f^{\otimes\mathbb{N}}$ on $\mathbb{R}^{\mathbb{N}}$ with its product Borel sigma-field. The space H indexes coefficients; the random coordinate sequence is not asserted to belong to H. Write $\mathcal{P}_\pm$ for the group of all signed coordinate permutations of H. For $T \in \mathrm{GL}(H)$, meaning bounded and boundedly invertible, let $\nu_T = (\Phi_T)_\#\mu$, where Φ_T is the measurable row-series realization constructed in lemma 3.6.

For any two probability measures define

$$\mathsf{A}(\lambda,\nu) := \int \sqrt{\frac{d\lambda}{dm}\frac{d\nu}{dm}}\,dm, \qquad h(\lambda,\nu)^2 := 2 - 2\mathsf{A}(\lambda,\nu), \tag{3.2}$$

where m dominates both measures. The value is independent of m. Let P_n select the first n coordinates and set

$$E_n(T) := -2\log\mathsf{A}\big((P_n)_\#\mu, (P_n)_\#\nu_T\big).$$

These are actual output marginals; they are not the laws obtained by replacing T with its principal matrix corner.

Theorem 3.1 (Complete i.i.d. criterion for arbitrary coordinate mixing)**.** *Under (3.1), for every $T \in \mathrm{GL}(H)$,*

$$\boxed{\begin{aligned} \nu_T \sim \mu \quad &\Longleftrightarrow \quad \exists Q \in \mathcal{P}_\pm :\ TQ^{-1} - I \in \mathcal{S}_2(H) \\ &\Longleftrightarrow \quad \sup_n E_n(T) < \infty. \end{aligned}} \tag{3.3}$$

The exact linear stabilizer is $\mathrm{Stab}(\mu) = \mathcal{P}_\pm$. *If these equivalent conditions fail, then $\nu_T \perp \mu$ and $E_n(T) \uparrow \infty$. On the equivalence branch, the projected likelihoods and reciprocal likelihoods are uniformly integrable in their respective reference measures. For sufficiently large n, the operators $I + P_n(TQ^{-1} - I)P_n$ are invertible. Their likelihoods converge to $d\nu_T/d\mu$ in $L^1(\mu)$, and their square roots converge in $L^2(\mu)$.*

The proof is given after the block and Fisher calculations. The dimension-free comparison in proposition 3.15 is a nonlinear estimate for Hellinger distance, with its natural saturation at large distances. Its local lower bound, applied to arbitrary tail marginals, is sufficient for necessity. No additive formula for the Hellinger energy of a mixed product law is used.

3.2 The exact block-product theorem

Let

$$X = \prod_{j \geq 1} \mathbb{R}^{d_j}, \qquad \mu = \bigotimes_{j \geq 1} \mu_j,$$

on the product Borel sigma-field. Assume that every μ_j has an everywhere positive Lebesgue density. For invertible matrices $T_j \in \mathrm{GL}(d_j, \mathbb{R})$, the coordinatewise map

$$T = \bigoplus_{j \geq 1} T_j$$

is measurable on X, and

$$T_\# \mu = \bigotimes_{j \geq 1} (T_j)_\# \mu_j.$$

Using (3.2), define the one-block affinity and action by

$$\rho_j(T_j) := \mathsf{A}\big(\mu_j, (T_j)_\# \mu_j\big), \qquad e_j(T_j) := -2 \log \rho_j(T_j). \tag{3.4}$$

The strict positivity of the block densities implies $0 < \rho_j(T_j) \leq 1$.

Theorem 3.2 (Exact block-product criterion)**.** *For every block-diagonal invertible T as above,*

$$T_{\#}\mu \sim \mu \quad \Longleftrightarrow \quad \sum_{j=1}^{\infty} e_j(T_j) < \infty. \tag{3.5}$$

If the series diverges, then $T_{\#}\mu \perp \mu$. More precisely, for every finite set $F \Subset \mathbb{N}$,

$$\mathsf{A}_F(T) = \prod_{j \in F} \rho_j(T_j), \qquad E_F(T) := -2 \log \mathsf{A}_F(T) = \sum_{j \in F} e_j(T_j), \tag{3.6}$$

and, whenever $F \subset G$,

$$E_G(T) - E_F(T) = \sum_{j \in G \setminus F} e_j(T_j). \tag{3.7}$$

Thus singularity is exactly the escape $E_F(T) \uparrow \infty$ along the directed family of finite coordinate sets.

Proof. Both formulas in (3.6) follow from tensorization of Hellinger affinity. Kakutani's theorem [6] says that two products of pairwise equivalent probability measures are equivalent precisely when their infinite product affinity is positive, and otherwise they are mutually singular. In the present notation,

$$\prod_j \rho_j(T_j) > 0 \quad \Longleftrightarrow \quad -\sum_j \log \rho_j(T_j) < \infty,$$

which is (3.5). The increment identity is immediate from additivity. □

The transformation parameter is not identifiable before quotienting by exact symmetries of the block laws. Put

$$H_j := \mathrm{Stab}(\mu_j) = \{Q \in \mathrm{GL}(d_j) : Q_{\#}\mu_j = \mu_j\}. \tag{3.8}$$

Then $e_j(T_j Q) = e_j(T_j)$ for $Q \in H_j$; the natural parameter is the right coset $T_j H_j$, not the matrix T_j itself.

We next state a useful Hilbert–Schmidt form of theorem 3.2. Its assumptions are deliberately explicit: they are the hypotheses which turn exact Hellinger energy into a quadratic matrix coordinate, rather than conclusions of Kakutani's theorem.

Corollary 3.3 (Uniform bounded blocks, modulo exact stabilizers)**.** *Assume the following.*

(B1) $\sup_j d_j < \infty$, *every* μ_j *is centered with identity covariance, and* H_j *is its exact linear stabilizer. Consequently* $H_j \subset O(d_j)$ *is compact.*

(B2) *The root-likelihood orbit*

$$\Phi_j : \mathrm{GL}(d_j)/H_j \longrightarrow L^2(\mu_j), \qquad \Phi_j(gH_j) = \sqrt{\frac{d(g_\# \mu_j)}{d\mu_j}},$$

is a topological embedding with closed image. Equip the quotient with the pullback metric

$$d_j^{\mathrm{root}}(gH_j, kH_j) := \|\Phi_j(gH_j) - \Phi_j(kH_j)\|_{L^2(\mu_j)}. \tag{3.9}$$

This metric is complete because its image is closed in $L^2(\mu_j)$.

(B3) *With*

$$r_j(gH_j) := \min_{Q \in H_j} \|gQ^{-1} - I\|_F,$$

there are constants $r_0, c, C > 0$, *independent of* j, *such that*

$$c\, r_j(gH_j)^2 \le e_j(g) \le C\, r_j(gH_j)^2 \quad \textit{whenever } r_j(gH_j) \le r_0. \tag{3.10}$$

This is the required uniform quadratic-mean differentiability and Fisher ellipticity assumption.

(B4) *The orbit action is uniformly proper at its maximum: for every* $\varepsilon > 0$,

$$\inf_j \inf_{r_j(gH_j) \ge \varepsilon} e_j(g) > 0. \tag{3.11}$$

Let $\mathcal{H} = \bigoplus_{j \ge 1}^2 \mathbb{R}^{d_j}$, *and suppose* $T = \bigoplus_j T_j$ *is bounded and boundedly invertible on* $\mathcal{H}$. *Then*

$$T_\# \mu \sim \mu \quad \Longleftrightarrow \quad \sum_j r_j(T_j H_j)^2 < \infty. \tag{3.12}$$

Equivalently, there is a block-orthogonal $Q = \bigoplus_j Q_j \in \prod_j H_j$ *such that*

$$TQ^{-1} - I \in \mathcal{S}_2(\mathcal{H}), \qquad \|TQ^{-1} - I\|_{\mathcal{S}_2}^2 = \sum_j \|T_j Q_j^{-1} - I\|_F^2 < \infty. \tag{3.13}$$

Proof. If $\sum_j e_j(T_j) < \infty$, then $e_j(T_j) \to 0$. Uniform properness forces $r_j(T_j H_j) \to 0$, after which the two bounds in (3.10) show that the two series are finite simultaneously. The reverse implication follows from the same local comparison, since only finitely many blocks lie outside the local chart. Compactness of H_j gives a minimizing Q_j in the definition of r_j. Since every Q_j is orthogonal, $Q = \bigoplus_j Q_j$ is a bounded orthogonal operator, and the last equivalence is the standard block formula for the Hilbert–Schmidt norm. The last assertion in (B2) concerns the explicitly defined one-block metric (3.9); it makes no completeness claim for an infinite product quotient. The class-preserving Cauchy structure for the latter is defined and proved complete in definition 4.6 and theorem 4.7. $\square$

The exact quantity in this corollary is the accumulated action $\sum_j e_j$, not a preassigned operator ideal. The Hilbert–Schmidt expression appears only after uniform DQM, ellipticity, and properness identify that action with squared Frobenius distance in the quotient chart.

3.3 An exact affine-coordinate theorem

Let $f > 0$ almost everywhere be a nondegenerate density on $\mathbb{R}$, put $g = \sqrt{f}$, and assume

$$g \in W^{1,2}(\mathbb{R}), \qquad xg' \in L^2(\mathbb{R}). \tag{3.14}$$

For $(u, v) \in \mathbb{R}^2$, let $q_{u,v}$ be the density of $e^u X + v$ when X has density f:

$$q_{u,v}(y) = e^{-u} f\big(e^{-u}(y - v)\big). \tag{3.15}$$

If $\psi = -f'/f$ denotes the location score, differentiation of the square root at the identity gives

$$D\sqrt{q}_{0,0}(u, v) = u\left(-\tfrac{1}{2}g - xg'\right) - vg', \qquad S_u(x) = x\psi(x) - 1, \quad S_v(x) = \psi(x). \tag{3.16}$$

Define

$$I_{\text{aff}} := \mathbb{E}_f \begin{bmatrix} S_u \\ S_v \end{bmatrix} \begin{bmatrix} S_u & S_v \end{bmatrix}, \qquad \delta(u, v) := -2 \log \mathsf{A}(f, q_{u,v}). \tag{3.17}$$

Theorem 3.4 (Diagonal dilation and translation)**.** *Assume* (3.14)*, assume that* I_{aff} *is positive definite, and assume that the affine root-density orbit is proper, in the concrete sense that*

$$\mathsf{A}(f, q_{u_n,v_n}) \longrightarrow 1 \quad \Longrightarrow \quad (u_n, v_n) \longrightarrow (0, 0). \tag{3.18}$$

For $\mu = f^{\otimes \mathbb{N}}$ and the coordinatewise affine map

$$T_{u,v}(x)_j = e^{u_j} x_j + v_j,$$

one has

$$(T_{u,v})_{\#}\mu \sim \mu \quad \Longleftrightarrow \quad u \in \ell^2 \text{ and } v \in \ell^2. \tag{3.19}$$

Otherwise the two product measures are mutually singular. In particular, for positive diagonal dilations,

$$\operatorname{diag}(a_j)_{\#}\mu \sim \mu \quad \Longleftrightarrow \quad (\log a_j)_j \in \ell^2. \tag{3.20}$$

Proof. Quadratic-mean differentiability and (3.16) give

$$\delta(u,v) = \frac{1}{4}(u,v) I_{\mathrm{aff}} (u,v)^{\mathsf{T}} + o(u^2 + v^2). \tag{3.21}$$

Positive definiteness supplies two-sided quadratic bounds near the origin. Condition (3.18) supplies a positive lower bound for δ outside every neighborhood of the origin. Hence

$$\sum_j \delta(u_j, v_j) < \infty \quad \Longleftrightarrow \quad \sum_j (u_j^2 + v_j^2) < \infty.$$

The result follows from theorem 3.2, with one-dimensional blocks. Setting $v = 0$ and $a_j = e^{u_j}$ gives (3.20). □

For an ℓ^2-valued realization, take $\mathrm{Law}((\sigma_j X_j)_j)$ with $\sigma \in \ell^2$ and $\mathbb{E}X_1^2 < \infty$, and use $x_j \mapsto e^{u_j} x_j + \sigma_j v_j$. Condition (3.19) then defines an affine homeomorphism of ℓ^2.

3.4 The Fisher operator for a general finite matrix

We first differentiate an arbitrary matrix acting in a finite block. The infinite-dimensional endpoint requires the nonlinear estimates proved below. Let

$$f(x) = Z^{-1} e^{-V(x)}$$

be an even, centered density of variance one. Put $s = V' = -f'/f$, and assume enough absolute continuity, boundary decay, and integrability to justify integration by parts and the identities

$$\mathbb{E}s(X) = 0, \qquad \mathbb{E}[s(X)X] = 1, \qquad J := \mathbb{E}[s(X)^2] < \infty, \qquad \mathbb{E}[s(X)^2 X^2] < \infty. \tag{3.22}$$

Let $X = (X_1, \dots, X_n) \sim f^{\otimes n}$, and consider $T_t = I + tA$, for a real $n \times n$ matrix A and sufficiently small t. Differentiating the Jacobian and the product density gives the score

$$S_A(X) = \sum_{i,j=1}^{n} A_{ij}\big(s(X_i)X_j - \delta_{ij}\big). \tag{3.23}$$

Set

$$\kappa := \mathbb{E}[(s(X)X - 1)^2]. \tag{3.24}$$

Theorem 3.5 (Spectral decomposition of the non-Gaussian Fisher operator)**.** *Equip $M_n(\mathbb{R})$ with the Frobenius inner product. Let P_{diag}, $P_{\mathrm{sym,off}}$, and P_{skew} be the orthogonal projections onto diagonal matrices, off-diagonal symmetric matrices, and skew-symmetric matrices, respectively. The Fisher quadratic form $\mathcal{I}_f^{(n)}(A) = \mathbb{E}[S_A^2]$ is represented by the self-adjoint operator*

$$\boxed{\mathscr{I}_f^{(n)} = \kappa P_{\mathrm{diag}} + (J+1)P_{\mathrm{sym,off}} + (J-1)P_{\mathrm{skew}}.} \tag{3.25}$$

The diagonal, off-diagonal symmetric, and skew-symmetric eigenspaces have, respectively, eigenvalues $\kappa, J+1, J-1$ and multiplicities

$$n, \qquad \frac{n(n-1)}{2}, \qquad \frac{n(n-1)}{2}. \tag{3.26}$$

Equivalently, writing

$$D = \operatorname{diag} A, \qquad S_0 = \frac{A + A^\top}{2} - D, \qquad K = \frac{A - A^\top}{2},$$

one has

$$\mathcal{I}_f^{(n)}(A) = \kappa\|D\|_F^2 + (J+1)\|S_0\|_F^2 + (J-1)\|K\|_F^2. \tag{3.27}$$

For the standard Gaussian, $J = 1$ and $\kappa = 2$, so the entire skew-symmetric subspace is the kernel. Under the stated regularity, $J \geq 1$, with equality only for the Gaussian. Hence, for a genuinely non-Gaussian f, if also $\kappa > 0$,

$$\min\{\kappa, J-1\}\|A\|_F^2 \leq \mathcal{I}_f^{(n)}(A) \leq \max\{\kappa, J+1\}\|A\|_F^2, \tag{3.28}$$

with constants independent of n.

Proof. Independence, parity, and (3.22) give

$$\mathbb{E}[S_A^2] = \kappa \sum_i A_{ii}^2 + \sum_{i<j} \left\{ J(A_{ij}^2 + A_{ji}^2) + 2A_{ij}A_{ji} \right\}. \tag{3.29}$$

For each $i < j$, put $S_{ij} = (A_{ij} + A_{ji})/2$ and $K_{ij} = (A_{ij} - A_{ji})/2$. The corresponding summand becomes

$$2(J+1)S_{ij}^2 + 2(J-1)K_{ij}^2.$$

Summing proves (3.27) and the claimed spectrum. Cauchy–Schwarz applied to $\mathbb{E}[s(X)X] = 1$ and $\mathbb{E}X^2 = 1$ gives $J \geq 1$; equality forces $s(X) = X$ almost surely and hence the standard Gaussian density. The final bounds follow by taking the smallest and largest of the three eigenvalues. $\square$

Gaussian orthogonal invariance makes every infinitesimal rotation invisible. For a genuinely non-Gaussian product density, the skew eigenvalue is $J - 1 > 0$, so the Fisher metric detects all matrix entries. This remains a tangent-space statement, not an infinite-dimensional endpoint criterion.

For rotations placed in independent two-dimensional blocks, the endpoint statement is again exact by theorem 3.2: the local block action is $e(R_\theta) = \frac{1}{2}(J-1)\theta^2 + o(\theta^2)$, and the global criterion is square summability of block distance to the exact rotation stabilizer.

3.5 Measurable linear realizations and likelihood closure

Throughout this subsection, f satisfies (3.1), $\mu = f^{\otimes \mathbb{N}}$ on $X = \mathbb{R}^{\mathbb{N}}$, and $\mu_n = f^{\otimes n}$. The sample sequence is not an ℓ^2-valued random variable. We therefore first specify how an operator on ℓ^2 acts on its law.

Lemma 3.6 (Measurable realization on each countable operator orbit)**.** *For every $T \in \mathcal{B}(\ell^2)$, there is a Borel map $\Phi_T : X \to X$ whose coordinates are the $L^2(\mu)$ row sums*

$$(\Phi_T x)_i = \sum_{j \geq 1} T_{ij} x_j.$$

These maps can be chosen so that the following statements hold. If a probability measure λ satisfies

$$\int \left| \sum_j a_j x_j \right|^2 d\lambda(x) \leq C_\lambda \|a\|_2^2 \quad \textit{for every finitely supported } a, \tag{3.30}$$

then $(\Phi_T)_{\#}\lambda$ also satisfies this condition, with constant at most $C_\lambda \|T\|^2$, and

$$\Phi_S \circ \Phi_T = \Phi_{ST} \quad \lambda\textit{-almost surely} \qquad (S, T \in \mathcal{B}(\ell^2)). \tag{3.31}$$

For each fixed countable subgroup $\Gamma \leq \mathrm{GL}(\ell^2)$, there is a Γ-invariant Borel set $D_\Gamma \subset X$ of full measure for every λ satisfying (3.30), on which these *restrictions form an action by bimeasurable automorphisms. In particular, this applies to μ and all its Γ-orbit laws, including singular orbit laws.*

Proof. For $a \in \ell^2$, the finite sums $\sum_{j \le n} a_j x_j$ have a unique $L^2(\lambda)$ limit L_a, by (3.30). Choose

$$n_k(a) = \min \left\{ n \ge k : \sum_{j>n} |a_j|^2 \le 2^{-4k} \right\}.$$

Define $\widehat{L}_a(x)$ to be the limit of $\sum_{j \le n_k(a)} a_j x_j$ when it exists, and zero otherwise. The estimate

$$\left\| \sum_{j \le n_k(a)} a_j x_j - L_a \right\|_{L^2(\lambda)}^2 \le C_\lambda 2^{-4k}$$

and Borel–Cantelli show that $\widehat{L}_a = L_a$, λ-almost surely. This one definition therefore works for every law satisfying (3.30); it does not depend on knowing whether that law is equivalent to μ. Set $\Phi_T(x)_i = \widehat{L}_{T^* e_i}(x)$. The construction is Borel, and the same is true jointly in the matrix coefficients of T and in x, since each tail norm and the integer choice n_k is Borel.

For finitely supported b, the finite sum of output coordinates equals $L_{T^* b}$ almost surely. This proves the asserted second-moment bound for the pushforward. For the i-th row of S, finite output sums satisfy

$$\sum_{j \le n} S_{ij} \widehat{L}_{T^* e_j} = L_{T^* P_n S^* e_i} \longrightarrow L_{T^* S^* e_i} \quad \text{in } L^2(\lambda).$$

The selected subsequence on the left also converges almost surely to $\Phi_S(\Phi_T(x))_i$, because the output law has the required second-moment bound. Uniqueness of the limit proves (3.31).

For a countable Γ, let B be the Borel set on which all its row limits, identity relations, and pairwise composition relations hold. The preceding argument gives $\lambda(B) = 1$ for every law under consideration. Put

$$D_\Gamma = \bigcap_{T \in \Gamma} \Phi_T^{-1}(B).$$

Every orbit law satisfies the same second-moment condition, so $\lambda(D_\Gamma) = 1$. The composition identities imply that D_Γ is invariant and that $\Phi_{T^{-1}}$ is the inverse of Φ_T there. This proves the final assertion. $\square$

Write $\nu_T = (\Phi_T)_{\#}\mu$. The preceding lemma makes assertions about a common carrier only for a fixed countable subgroup; no single pointwise action of all of $\mathrm{GL}(\ell^2)$ on a common conull set is required. The subsequences in its proof are also essential: a covariance bound gives convergence of arbitrary row sums in L^2, but does not by itself give almost-sure convergence of all ordinary partial sums under a correlated orbit law. On the original independent product law, the versions agree with the usual almost-sure row sums.

Write $D(\lambda\|\nu) = \int \log(d\lambda/d\nu)\, d\lambda$ for relative entropy, with value $+\infty$ when absolute continuity fails.

Lemma 3.7 (A dimension-free entropy upper bound)**.** *Let $L = \|V''\|_\infty$. For every $A \in \mathrm{GL}(n,\mathbb{R})$ with smallest singular value at least $m > 0$,*

$$D(A_{\#}\mu_n\|\mu_n) \leq C_L(m)\|A - I\|_F^2, \qquad C_L(m) = \frac{L}{2} + \frac{1}{2\min\{1,m\}^2}. \tag{3.32}$$

The constant is independent of n. Moreover,

$$D(\mu_n\|A_{\#}\mu_n) = D((A^{-1})_{\#}\mu_n\|\mu_n). \tag{3.33}$$

Proof. Put $K = A - I$ and $s = V'$. The standing assumptions justify integration by parts and give $\mathbb{E}[s(X_i)X_j] = \delta_{ij}$. In particular, bounded V'', evenness, and finite second moment make the relevant integrands integrable; cutoff integration by parts gives the diagonal identity. The change-of-variables formula and Taylor's upper bound yield

$$\begin{aligned} D(A_{\#}\mu_n\|\mu_n) &= \mathbb{E}\sum_i \{V((AX)_i) - V(X_i)\} - \log|\det A| \\ &\leq \operatorname{tr} K - \log|\det A| + \frac{L}{2}\|K\|_F^2. \end{aligned}$$

If σ_i are the singular values of A, then $\operatorname{tr} A \leq \sum_i \sigma_i$ and

$$\sum_i (\sigma_i - 1)^2 \leq \|A - I\|_F^2.$$

The latter follows by expanding both sides and using the former trace inequality. Taylor's formula for $u - 1 - \log u$ gives

$$u - 1 - \log u \leq \frac{(u-1)^2}{2\min\{1,m\}^2} \qquad (u \geq m).$$

Consequently,

$$\operatorname{tr} K - \log|\det A| \le \sum_i (\sigma_i - 1 - \log \sigma_i) \le \frac{\|K\|_F^2}{2\min\{1,m\}^2},$$

which proves (3.32). Applying the invertible map A^{-1} to both arguments of relative entropy proves (3.33). $\square$

Proposition 3.8 (Hilbert–Schmidt sufficiency and RN closure)**.** *Suppose $T = I + K \in \mathrm{GL}(\ell^2)$ and $K \in \mathcal{S}_2$. Then $\nu_T \sim \mu$. Set $T_n = I + P_n K P_n$, discarding the finitely many terms before these operators become invertible. Their densities $Z_n = d\nu_{T_n}/d\mu$ converge in $L^1(\mu)$ to the strictly positive density $Z_T = d\nu_T/d\mu$, and their square roots converge in $L^2(\mu)$. The inverse approximants have the analogous closure:*

$$\frac{d\nu_{T_n^{-1}}}{d\mu} \longrightarrow \frac{d\nu_{T^{-1}}}{d\mu} \quad \text{in } L^1(\mu).$$

Both $D(\nu_T\|\mu)$ and $D(\mu\|\nu_T)$ are finite. If $\nu_S \sim \mu$ and $\nu_T \sim \mu$, the RN derivatives obey

$$Z_{ST}(y) = Z_S(y) Z_T(\Phi_{S^{-1}} y), \tag{3.34}$$

$$Z_T(y) Z_{T^{-1}}(\Phi_{T^{-1}} y) = 1. \tag{3.35}$$

If $Q \in \mathrm{GL}(\ell^2)$ is an exact linear symmetry of μ, the same equivalence conclusion holds whenever $TQ^{-1} - I \in \mathcal{S}_2$ and T is boundedly invertible.

Proof. Since $P_n K P_n \to K$ in Hilbert–Schmidt norm, $T_n \to T$ in operator norm. Thus T_n is eventually invertible and both $\|T_n\|$ and $\|T_n^{-1}\|$ are uniformly bounded. If A_n denotes its first n-coordinate block, its density is the explicit finite-dimensional likelihood

$$Z_n(x) = |\det A_n|^{-1} \prod_{i=1}^n \frac{f((A_n^{-1} x_{1:n})_i)}{f(x_i)}. \tag{3.36}$$

These approximant densities need not form a martingale.

The operators $T_n^{-1} T_m$ act in a finite coordinate block and have uniformly bounded norms and inverses. The entropy lemma and finite change of variables therefore give, with a constant independent of m, n,

$$D(\nu_{T_m}\|\nu_{T_n}) = D(\nu_{T_n^{-1}T_m}\|\mu) \le C\|T_m - T_n\|_{\mathcal{S}_2}^2, \tag{3.37}$$

$$\|Z_m - Z_n\|_{L^1(\mu)} \le \sqrt{2C}\|T_m - T_n\|_{\mathcal{S}_2}. \tag{3.38}$$

The second line is Pinsker's inequality. Hence $Z_n \to Z$ in $L^1(\mu)$, with $Z \geq 0$ and $\int Z\,d\mu = 1$. For each fixed finite set of output coordinates, $\Phi_{T_n} X$ converges to $\Phi_T X$ in L^2, by the row construction. Bounded continuous cylinder tests consequently identify $Z\mu = \nu_T$. In particular, this limit is the law of the stated operator and cannot be an unrelated limiting density. The inequality $(\sqrt{a} - \sqrt{b})^2 \leq |a - b|$ gives convergence of square roots.

The same argument applies to the finite-block operators T_n^{-1}:

$$\|T_n^{-1} - T^{-1}\|_{\mathcal{S}_2} \leq \|T_n^{-1}\|\,\|T_n - T\|_{\mathcal{S}_2}\,\|T^{-1}\| \longrightarrow 0.$$

It identifies their L^1 density limit as $\nu_{T^{-1}}$, which is therefore absolutely continuous with respect to μ. On the common carrier for the countable group generated by T, T_n, lemma 3.6 gives the inverse relations independently of any absolute-continuity assertion. Thus

$$\mu = (\Phi_T)_{\#}\nu_{T^{-1}} \ll (\Phi_T)_{\#}\mu = \nu_T.$$

This proves equivalence and strict positivity of Z.

The finite entropy lemma bounds $D(\nu_{T_n}\|\mu)$ and $D(\mu\|\nu_{T_n})$ uniformly. Along an almost-surely convergent subsequence of the densities, Fatou's lemma applied respectively to $z\log z - z + 1$ and $z - 1 - \log z$ proves finiteness of both limiting entropies. Change of variables on the common carrier proves (3.34); the case $S = T$, with the second factor indexed by T^{-1}, gives (3.35). Finally, $T = (TQ^{-1})Q$ and $\nu_Q = \mu$ give the last assertion. $\square$

Lemma 3.9 (Dichotomy for every bounded invertible orbit)**.** *For every $T \in \mathrm{GL}(\ell^2)$,*

$$\nu_T \sim \mu \quad \text{or} \quad \nu_T \perp \mu. \tag{3.39}$$

This conclusion uses only positivity of f, its centered finite second moment, and $\sqrt{f} \in W^{1,2}(\mathbb{R})$; in particular, it holds under (3.1).

Proof. Put $g = \sqrt{f}$. The translation estimate

$$1 - \mathsf{A}(f, f(\cdot - a)) = \tfrac{1}{2}\|g(\cdot - a) - g\|_2^2 \leq \tfrac{1}{2}\|g'\|_2^2 a^2$$

and Kakutani's theorem imply $(\tau_h)_{\#}\mu \sim \mu$ for every $h \in \ell^2$. Under the standing assumptions, $g' = -(V'/2)g \in L^2$, since $|V'(x)| \leq L|x|$. The row realizations satisfy

$$\Phi_T(x + T^{-1}h) = \Phi_T(x) + h \quad \mu\text{-almost surely;}$$

this follows directly by adding the absolutely convergent scalar row pairing with the ℓ^2 vector $T^{-1}h$. Consequently, ν_T is also quasi-invariant under every ℓ^2 translation.

Let H be the countable additive group generated by $\{qe_j, qTe_j : q \in \mathbb{Q},\ j \geq 1\}$. Both μ and ν_T are quasi-invariant and ergodic under translation by H. Indeed, an event invariant modulo μ under all rational finite-coordinate translations has, by positivity of the finite-dimensional densities, almost-everywhere constant sections in each finite coordinate block. It is therefore a tail event modulo μ, and has probability zero or one. Here one uses the elementary fact that a measurable function on $\mathbb{R}^n$ invariant almost everywhere under rational translations is constant almost everywhere, obtained, for example, by convolution with smooth approximate identities. For ν_T, pull an invariant event back by Φ_T; invariance under qTe_j becomes invariance under qe_j, giving the same zero–one conclusion.

Write $\nu_T = \nu_a + \nu_s$ for its Lebesgue decomposition relative to μ. Choose a Borel μ-null set N_0 supporting ν_s, and let $N = \bigcup_{h\in H}(N_0 + h)$. Quasi-invariance of μ makes N a μ-null set; it is exactly H-invariant and still supports ν_s. Thus $\nu_T(N) = \nu_s(X)$, and ergodicity of ν_T makes this number zero or one. The latter case gives singularity. In the former case, write $\nu_T = Z\mu$. Quasi-invariance of both measures implies that $\{Z > 0\}$ is invariant modulo μ under H. Its μ-measure is positive because $\int Z\,d\mu = 1$, and ergodicity of μ makes that measure one. This gives equivalence. $\square$

Lemma 3.10 (Projected likelihoods and two-sided uniform integrability)**.** *For $T \in \mathrm{GL}(\ell^2)$, every finite output law $\nu_{T,N} = (P_N)_{\#}\nu_T$ is equivalent to μ_N. Writing $\mathcal{F}_N = \sigma(x_1, \dots, x_N)$, define*

$$R_N(x) = \frac{d\nu_{T,N}}{d\mu_N}(x_{1:N}), \qquad \mathsf{A}_N(T) = \mathbb{E}_\mu\sqrt{R_N}, \qquad E_N(T) = -2\log \mathsf{A}_N(T). \tag{3.40}$$

Then (R_N) is a nonnegative mean-one μ-martingale and

$$\mathsf{A}_N(T) \downarrow \mathsf{A}(\mu, \nu_T), \qquad \nu_T \sim \mu \iff \sup_N E_N(T) < \infty. \tag{3.41}$$

Otherwise $E_N(T) \uparrow \infty$ and $\nu_T \perp \mu$. If $\nu_T \sim \mu$ with density Z_T, then

$$R_N = \mathbb{E}_\mu[Z_T \mid \mathcal{F}_N] \longrightarrow Z_T \qquad \text{in } L^1(\mu), \tag{3.42}$$

$$R_N^{-1} = \mathbb{E}_{\nu_T}[Z_T^{-1} \mid \mathcal{F}_N] \longrightarrow Z_T^{-1} \qquad \text{in } L^1(\nu_T). \tag{3.43}$$

In particular, both likelihood martingales are uniformly integrable in their respective reference measures. In the Hilbert–Schmidt case of proposition 3.8, *they additionally have uniformly bounded forward and reverse relative entropies.*

Proof. The first N rows of T are linearly independent, since T^* is injective. Some N input columns therefore form an invertible $N \times N$ minor A. Split the input into these coordinates and their complement. The first N outputs have the form $AU + W$, where $U \sim \mu_N$ and W is independent of U. Conditioning on W writes their density as a mixture of translates of the strictly positive density of AU. This mixture is positive and finite almost everywhere, proving the finite-dimensional equivalence.

Consistency of projected laws proves the martingale assertion. The martingale limit is $R_\infty = d(\nu_T)_a/d\mu$, the density of the absolutely continuous part of ν_T. To see the identification explicitly, Fatou's lemma on cylinder sets gives $R_\infty \mu \leq \nu_T$, first on the cylinder algebra and then on the generated sigma-field. Conversely, if $(\nu_T)_a = Z\mu$, then $\mathbb{E}_\mu[Z \mid \mathcal{F}_N] \leq R_N$, so martingale convergence gives $Z \leq R_\infty$. The two inequalities identify the limit. Since $\mathbb{E}_\mu(\sqrt{R_N})^2 = 1$, the square roots are uniformly integrable in $L^1(\mu)$. They thus converge in L^1 to $\sqrt{R_\infty}$. Conditional Jensen's inequality gives the monotonicity of their expectations, and consequently

$$\lim_N \mathsf{A}_N(T) = \int \sqrt{R_\infty}\, d\mu = \mathsf{A}(\mu, \nu_T).$$

The dichotomy in lemma 3.9 now proves (3.41) and the singular alternative.

Under equivalence, the forward conditional-expectation identity is the definition of a projected density. The reverse identity follows because for $B \in \mathcal{F}_N$,

$$\int_B R_N^{-1}\, d\nu_T = \mu(B) = \int_B Z_T^{-1}\, d\nu_T.$$

Conditional-expectation convergence gives both L^1 limits and hence both uniform-integrability assertions. The entropy bounds in the Hilbert–Schmidt case follow from proposition 3.8 by conditional Jensen's inequality, or equivalently by contraction of relative entropy under projection. $\square$

3.6 Dimension-free nonlinear Hellinger coercivity

We use the Hellinger convention (3.2). All constants below are independent of both the number of observed coordinates and the number of source coordinates. The source may be countably infinite. Throughout this subsection, X_j are independent with an even, non-Gaussian density f, mean zero, variance one, and all moments finite. For the final matrix comparison we use the entropy estimate (3.32), proved from (3.1).

Lemma 3.11 (A Hellinger detector inequality)**.** *For probability measures P, Q and a real statistic F square integrable under both measures, put $d = \mathbb{E}_P F - \mathbb{E}_Q F$. Then*

$$(1 - h(P,Q)^2)d^2 \le 2h(P,Q)^2\{\operatorname{Var}_P F + \operatorname{Var}_Q F\}. \tag{3.44}$$

In particular, if $h(P,Q)^2 \le 1/2$, then $d^2 \le 4h(P,Q)^2(\operatorname{Var}_P F + \operatorname{Var}_Q F)$.

Proof. Take a common dominating measure, write its densities as p, q, and set $c = (\mathbb{E}_P F + \mathbb{E}_Q F)/2$. Cauchy–Schwarz and $(\sqrt{p} + \sqrt{q})^2 \le 2(p+q)$ give

$$d^2 \le h(P,Q)^2 \int (F-c)^2(\sqrt{p}+\sqrt{q})^2 \le 2h(P,Q)^2 \left\{\operatorname{Var}_P F + \operatorname{Var}_Q F + \frac{d^2}{2}\right\}.$$

Rearrange. □

Lemma 3.12 (Covariance coercivity for arbitrary row maps)**.** *Let $A : \ell^2(J) \to \mathbb{R}^n$ be a bounded row map, where J is finite or countable, with $\|A\|_{\mathrm{op}} \le b$. Define AX by the L^2 limits of its coordinate series, put $\Sigma = AA^\mathsf{T}$, and set $h = h(f^{\otimes n}, \mathrm{Law}(AX))$. There is a constant $C_{f,b} < \infty$ such that*

$$(1 - h^2)\|\Sigma - I_n\|_F^2 \le C_{f,b} h^2. \tag{3.45}$$

Proof. For a symmetric matrix B and independent standardized coordinates,

$$\operatorname{Var}(X^\mathsf{T} BX) = 2\|B\|_F^2 + (\mathbb{E}X_1^4 - 3)\sum_j B_{jj}^2 \le c_f\|B\|_F^2, \qquad c_f = 2 + |\mathbb{E}X_1^4 - 3|.$$

The same bound holds for a self-adjoint Hilbert–Schmidt matrix on a countable source, with the quadratic form centered and defined in L^2; this follows by truncation. Apply this to the statistic $F(y) = y^\mathsf{T} Dy$, where $D = \Sigma - I_n$. Its expectation difference under the two laws is $\|D\|_F^2$, while its two variances are bounded by $c_f\|D\|_F^2$ and $c_f b^4\|D\|_F^2$, respectively. The latter bound uses $\|A^\mathsf{T} DA\|_{\mathcal{S}_2} \le b^2\|D\|_F$. Apply lemma 3.11 and divide by $\|D\|_F^2$ when this quantity is nonzero. □

The first non-Gaussian moment gives a finite-degree detector for rotations. Let $r \ge 4$ be the least degree at which the moments of f differ from those of the standard Gaussian. Such an r exists because the Gaussian moment sequence determines its probability law; evenness makes r even. Write H_k for the probabilists' Hermite polynomial, defined by

$$e^{tx - t^2/2} = \sum_{k \ge 0} H_k(x)\frac{t^k}{k!}, \qquad \gamma_r := \mathbb{E}H_r(X_1) \ne 0.$$

Thus $\mathbb{E}H_k(X_1) = 0$ for $1 \le k < r$, and γ_r is also the first nonzero non-Gaussian cumulant. Only the moments through order $2r$ enter our estimates.

For a symmetric tensor $B \in (\mathbb{R}^n)^{\odot r}$ define its Gaussian Wick polynomial by

$$W_B(x) := \sum_{|\alpha|=r} \frac{r!}{\alpha!} B_\alpha \prod_{i:\alpha_i>0} H_{\alpha_i}(x_i), \qquad \|B\|^2 = \sum_{|\alpha|=r} \frac{r!}{\alpha!} |B_\alpha|^2. \tag{3.46}$$

Here B_α is the common tensor entry with multiplicity pattern α, and the tensor norm is the ordinary Hilbert tensor norm.

Lemma 3.13 (A dimension-free Wick variance bound)**.** *There is $C_{f,r} < \infty$ such that, for every finite-dimensional symmetric r-tensor B,*

$$\mathbb{E}W_B(X) = \gamma_r \sum_i B_{i\cdots i}, \qquad \operatorname{Var} W_B(X) \le C_{f,r}\|B\|^2. \tag{3.47}$$

If $A : \ell^2(J) \to \mathbb{R}^n$ is a coisometry ($AA^\mathsf{T} = I_n$), the same variance bound holds for $W_B(AX)$. Its expectation is

$$\mathbb{E}W_B(AX) = \gamma_r \langle B, C_A \rangle, \qquad C_A := \sum_{j\in J} a_j^{\otimes r}, \quad a_j = Ae_j. \tag{3.48}$$

Proof. Group the terms in (3.46) by their coordinate support. For a support containing at least two indices, each positive degree is strictly less than r, so the corresponding Hermite factor has mean zero. For a singleton support replace H_r by $H_r - \gamma_r$ when centering. Terms with different supports are therefore orthogonal in $L^2(f^{\otimes n})$. For a fixed support there are at most 2^{r-1} positive compositions of r. Cauchy–Schwarz within each such group, the finite moments through order $2r$, and $r!/\alpha! \le r!$ bound its variance by a constant times its contribution to $\|B\|^2$. Summing proves (3.47), including the expectation formula.

The generating function for the Wick polynomial gives the tensor identity

$$W_B(AX) = W_{(A^\mathsf{T})^{\otimes r}B}(X) \quad \text{when } AA^\mathsf{T} = I_n.$$

Since A^T is an isometry, the pulled-back tensor has norm $\|B\|$. This proves the assertion for a finite source. For a countable source, expand first in finitely many source coordinates and pass to the limit in the centered Wick expansion. The already proved variance bound makes this expansion continuous in Hilbert tensor norm. Equivalently, one may truncate the coordinate series for AX, using

$$\mathbb{E}\left|\sum_j a_j X_j\right|^{2r} \le C_{f,r}\left(\sum_j |a_j|^2\right)^r \qquad \text{for finitely supported } a.$$

To prove this bound, expand the even power: independence and evenness remove every term with an odd multiplicity, and the remaining multiplicity patterns are bounded by a constant depending only on the moments through $2r$ times the expansion of $(\sum_j |a_j|^2)^r$. Applying the bound to differences of truncations proves convergence in L^{2r}. In this truncation the Gaussian covariance correction also converges to I_n. The expectation series is absolutely convergent: each tensor coefficient is a finite linear combination of sums $\sum_j \prod_{k=1}^r A_{i_k j}$, which converge by Hölder and the ℓ^2 row bounds. This proves (3.48) for a countable source as well. ☐

Lemma 3.14 (Dimension-free moment detectors)**.** *For every bounded row map $A : \ell^2(J) \to \mathbb{R}^n$, the covariance estimate (3.45) holds. If additionally $AA^\mathsf{T} = I_n$, put*

$$D_n := \sum_{i=1}^n e_i^{\otimes r}, \qquad \Delta_A := D_n - C_A, \qquad h = h(f^{\otimes n}, \mathrm{Law}(AX)).$$

Then

$$(1-h^2)\|\Delta_A\|^2 \le C_{f,r} h^2. \tag{3.49}$$

In particular, near-maximal affinity forces the entire cumulant defect to zero in Hilbert tensor norm, uniformly in both dimensions.

Proof. The covariance assertion is lemma 3.12. For the second assertion, use $F = W_{\Delta_A}$ in lemma 3.11. Its expectation difference is $\gamma_r\|\Delta_A\|^2$, and lemma 3.13 bounds each variance by $C_{f,r}\|\Delta_A\|^2$. Divide by the latter squared tensor norm if nonzero, absorbing γ_r^{-2} into the constant. ☐

Proposition 3.15 (Finite nonlinear coercivity modulo signed permutations)**.** *Let $\mathfrak{P}_n$ be the signed permutation matrices. Under (3.1), fix $0 < a \le b < \infty$. There are $c, C > 0$, depending only on f, a, b, such that for every n and every $T \in \mathrm{GL}(n,\mathbb{R})$ with singular values in $[a,b]$,*

$$c\min\{d_n(T)^2, 1\} \le h(f^{\otimes n}, T_\# f^{\otimes n})^2 \le C\min\{d_n(T)^2, 1\}, \qquad d_n(T) := \min_{Q\in\mathfrak{P}_n} \|T-Q\|_F. \tag{3.50}$$

These are nonlinear estimates uniform in dimension; they do not assert an additive formula for marginal Hellinger increments.

Proof. First let $T = O$ be orthogonal. Orthogonality of the tensors $(Oe_j)^{\otimes r}$ gives

$$\|\Delta_O\|^2 = 2\left(n - \sum_{i,j} |O_{ij}|^r\right).$$

The matrix $p_{ij} = O_{ij}^2$ is doubly stochastic. Its decomposition into permutation matrices implies that some permutation π satisfies

$$\sum_i p_{i,\pi(i)} \geq \sum_{i,j} p_{ij}^2 \geq \sum_{i,j} |O_{ij}|^r.$$

Choosing the signs of its permutation matrix to agree with O therefore shows

$$d_n(O)^2 \leq 2\left(n - \sum_i |O_{i,\pi(i)}|\right) \leq \|\Delta_O\|^2. \tag{3.51}$$

Consequently lemma 3.14 gives $d_n(O) \leq C_f h(f^{\otimes n}, O_\# f^{\otimes n})$ whenever the latter Hellinger distance is at most $1/\sqrt{2}$.

For general T write its left polar decomposition as $T = BO$, with $B = (TT^\top)^{1/2}$. Put $h_T = h(f^{\otimes n}, T_\# f^{\otimes n})$. If $h_T \leq 1/\sqrt{2}$, lemma 3.12 gives

$$\|B - I\|_F \leq \frac{\|TT^\top - I\|_F}{a+1} \leq C_{f,a,b} h_T.$$

Hellinger invariance under $O^\top$, the elementary inequality $h(P,Q)^2 \leq D(Q\|P)$, and (3.32) imply

$$h(T_\# f^{\otimes n}, O_\# f^{\otimes n}) = h((O^\top BO)_\# f^{\otimes n}, f^{\otimes n}) \leq C_{f,a}\|B - I\|_F.$$

The triangle inequality now bounds the Hellinger distance of O from the reference law by $C_{f,a,b}h_T$. For sufficiently small h_T, the orthogonal case applies, and

$$d_n(T) \leq \|T - O\|_F + d_n(O) \leq C_{f,a,b} h_T.$$

If h_T is not this small, the lower bound in (3.50) follows by decreasing its constant, since $\min\{d_n(T)^2, 1\} \leq 1$.

Finally choose a minimizing signed permutation Q. Its action preserves $f^{\otimes n}$, so the entropy bound applied to $TQ^\top$ yields $h_T^2 \leq C_{f,a}\|TQ^\top - I\|_F^2 = C_{f,a} d_n(T)^2$. Combine this with $h_T^2 \leq 2$ to obtain the upper bound. $\square$

3.7 Tail rigidity and the endpoint theorem

Lemma 3.16 (Equivalent laws have asymptotically identical tails)**.** *If $\nu \sim \mu$, then*

$$\inf_{F \Subset \{N+1,N+2,\ldots\}} \mathsf{A}\big((P_F)_{\#}\mu, (P_F)_{\#}\nu\big) \longrightarrow 1 \qquad (N \to \infty). \tag{3.52}$$

Here P_F denotes projection onto an arbitrary finite coordinate set.

Proof. Let $p = d\nu/d\mu$ and $\mathcal{G}_N = \sigma(x_i : i > N)$. The density of the restriction of ν to $\mathcal{G}_N$, relative to that of μ, is $p_N = \mathbb{E}_\mu[p \mid \mathcal{G}_N]$. The reverse martingale theorem and triviality of the product tail give $p_N \to 1$ in $L^1(\mu)$. Hence

$$\mathbb{E}_\mu(\sqrt{p_N} - 1)^2 \le \mathbb{E}_\mu|p_N - 1| \longrightarrow 0.$$

Hellinger distance decreases under measurable projection, uniformly for all finite $F \subset \{N+1, N+2, \ldots\}$, proving the assertion. ☐

Lemma 3.17 (Covariance rigidity)**.** *If $T_{\#}\mu \sim \mu$, then $TT^* - I \in \mathcal{S}_2$.*

Proof. Let $C = TT^* - I$. By lemma 3.16, all finite-coordinate affinities sufficiently far out in the tail are at least $3/4$, so the squared Hellinger distances in lemma 3.14 are at most $1/2$. The dimension-free estimate (3.45) consequently bounds $\|P_F C P_F\|_F$, uniformly over every finite set F in that tail. Increasing F to the whole tail and summing squared matrix entries shows that $P_{>N} C P_{>N}$ is Hilbert–Schmidt. The remaining rows and columns form a bounded finite-rank operator, so C is Hilbert–Schmidt as well. ☐

Retain the first non-Gaussian order r and coefficient γ_r from lemma 3.13. For an orthogonal operator $U = (u_{ij})$, define its order-r defect array by

$$D_{i_1 \ldots i_r}(U) := \mathbf{1}_{\{i_1 = \cdots = i_r\}} - \sum_j u_{i_1 j} \cdots u_{i_r j}. \tag{3.53}$$

The series converges absolutely by Hölder's inequality and the ℓ^2 bounds on the rows. The next argument explicitly controls the indices omitted by a coordinate tail projection.

Lemma 3.18 (Tensor rigidity and a single signed permutation)**.** *If U is orthogonal and $U_{\#}\mu \sim \mu$, then*

$$\sum_i \left(1 - \sum_j |u_{ij}|^r\right) < \infty, \qquad U - Q \in \mathcal{S}_2 \quad \textit{for some signed permutation } Q. \tag{3.54}$$

Proof. Apply lemma 3.16 and the dimension-free estimate (3.49) to every finite coordinate set in a sufficiently far tail. The squared norms of the corresponding restrictions of $D(U)$ are uniformly bounded. Monotone convergence of the sums of squared entries therefore shows that the restriction of $D(U)$ to $\{N+1, N+2, \dots\}^r$ is square summable.

Write $u_j = Ue_j$. For each fixed first index i, the tensor series

$$\sum_j u_{ij} u_j^{\otimes(r-1)}$$

converges in the Hilbert tensor product: the tensors $u_j^{\otimes(r-1)}$ are orthonormal and $\sum_j u_{ij}^2 = 1$. The squared norm of this slice of the defect is exactly

$$\left\| e_i^{\otimes(r-1)} - \sum_j u_{ij} u_j^{\otimes(r-1)} \right\|^2 = 2\left(1 - \sum_j |u_{ij}|^r\right) \le 2. \tag{3.55}$$

By symmetry, the part of the array with at least one index at most N has squared norm at most $2rN$. Adding these finitely many strips proves that the full array is square summable. Summing (3.55) gives the first assertion.

Put $d_i = 1 - \sum_j |u_{ij}|^r$ and $m_i = \max_j |u_{ij}|$. Since $r \ge 4$,

$$\sum_j |u_{ij}|^r \le m_i^{r-2} \le m_i^2, \qquad 1 - m_i^2 \le d_i.$$

Outside a finite initial set of rows, $d_i < 1/2$. Each such row has a unique entry of squared modulus greater than $1/2$, in a column $\pi(i)$. No two of these rows choose the same column, because every column has squared norm one. With $\varepsilon_i = \operatorname{sgn}(u_{i,\pi(i)})$,

$$\sum_{i>N} \left\| U^* e_i - \varepsilon_i e_{\pi(i)} \right\|^2 = 2 \sum_{i>N} (1 - m_i) \le 2 \sum_{i>N} d_i < \infty.$$

Define Q_0 by these signed rows on $i > N$ and by zero rows on $i \le N$. Then $U - Q_0$ is Hilbert–Schmidt. A compact perturbation of an invertible operator is Fredholm of index zero, so Q_0 has index zero. Its cokernel is precisely the first N output coordinates; its kernel is the span of the unused input coordinates. Consequently there are exactly N unused columns. Match them bijectively to the first N rows and choose any signs there. This completes Q_0 to a single signed permutation Q, with $U - Q \in \mathcal{S}_2$. □

Proposition 3.19 (Exact linear stabilizer)**.** *The bounded invertible linear stabilizer of μ is exactly the group of signed coordinate permutations.*

Proof. Evenness and identical distribution make every signed permutation a symmetry. Conversely, if $T_{\#}\mu = \mu$, covariance gives $TT^* = I$, so T is orthogonal. The moment assumptions below order r imply, for each row of this orthogonal operator,

$$\mathbb{E}\left(\sum_j t_{ij}X_j\right)^r = (r-1)!! + \gamma_r \sum_j |t_{ij}|^r.$$

The same moment under μ is $(r-1)!! + \gamma_r$, so $\sum_j |t_{ij}|^r = 1$. A unit ℓ^2 vector has this property for $r > 2$ only when exactly one coordinate has modulus one. Orthogonality then makes the selected coordinates a bijection. Thus T is a signed permutation. The moment identity for infinite rows follows by truncation in L^r; the finite r-th moment and the standard moment bound for sums of independent centered variables justify that convergence. □

Proof of theorem 3.1. Suppose first that $T_{\#}\mu \sim \mu$. By lemma 3.17, $C = TT^* - I$ is Hilbert–Schmidt. The positive invertible operator $S = (TT^*)^{1/2}$ satisfies

$$S - I = (TT^* - I)(S + I)^{-1} \in \mathcal{S}_2.$$

The Hilbert–Schmidt sufficiency theorem proposition 3.8 applies to S and S^{-1}. Thus, for the orthogonal operator $U = S^{-1}T$, measurable composition and preservation of absolute continuity under pushforward give

$$U_{\#}\mu = (S^{-1})_{\#}(T_{\#}\mu) \sim (S^{-1})_{\#}\mu \sim \mu.$$

By lemma 3.18, $U - Q \in \mathcal{S}_2$ for a signed permutation Q. Therefore

$$T - Q = (S - I)U + (U - Q) \in \mathcal{S}_2.$$

Conversely, if $T - Q \in \mathcal{S}_2$ for a signed permutation Q, then $TQ^{-1} = I + (T-Q)Q^{-1}$ is boundedly invertible and differs from the identity by a Hilbert–Schmidt operator. Since $Q_{\#}\mu = \mu$, proposition 3.8 gives $T_{\#}\mu \sim \mu$. If no such Q exists, equivalence is impossible, and lemma 3.9 gives mutual singularity. Also, proposition 3.19 identifies the quotient symmetry group appearing in the criterion. The energy equivalence and its divergent alternative follow from lemma 3.10. Applying proposition 3.8 to TQ^{-1} identifies the principal-truncation likelihood limit with $d\nu_T/d\mu$, since $\nu_{TQ^{-1}} = \nu_T$. Finally, (3.42)–(3.43) give both uniform-integrability assertions. □

3.8 Moment assumptions and scope

Only moments through order $2r$ are used by the detector argument. Accordingly, the all-moments assumption can be replaced by the existence of an even $r \geq 4$ such that the moments below r agree with the standard Gaussian moments, the moment of order r differs, and $\mathbb{E}|X|^{2r} < \infty$.

The bounded second derivative in (3.1) supplies the dimension-free entropy upper bound; the first non-Gaussian moment supplies the nonlinear lower bound. These play different roles and are both verified above. The result covers every bounded invertible cross-coordinate operator for this density class. It makes no unrestricted claim for densities without the stated regularity or moment conditions. In the Gaussian case the moment detector vanishes at every order, and the exact stabilizer enlarges to the orthogonal group, as in section 2.

4 Projection increments and class-preserving completion

The preceding product calculation suggests an order of construction that does not mention an operator ideal. One first asks what every finite observation can see, then records the information added by refinement, and only afterwards completes the finite-dimensional core. This order is important outside the Gaussian chart: an infinite-dimensional norm chosen in advance can confuse a feature of one parametrization with an invariant of the underlying laws.

Throughout this section, $(X, \mathcal{A})$ is a standard Borel space, $\mathcal{G}$ is a group of bimeasurable automorphisms of X, and μ_0 is a probability law on X. We write

$$\mathcal{O}(\mu_0) := \{g_{\#}\mu_0 : g \in \mathcal{G}\}.$$

Linearity of X or of the action will not be used until a Hilbert information chart is introduced. Thus the measure-theoretic construction below applies to arbitrary automorphism orbits.

4.1 Finite tests and joint observations

Let $\mathscr{T}$ be a family of measurable maps $\tau : X \to X_\tau$, where each X_τ is a finite-dimensional standard Borel space. For a finite set $F \Subset \mathscr{T}$, put

$$Q_F : X \longrightarrow \prod_{\tau \in F} X_\tau, \qquad Q_F(x) := (\tau(x))_{\tau \in F}, \qquad \mathcal{A}_F := \sigma(Q_F).$$

If $F \subset F'$, then Q_F is a measurable factor of $Q_{F'}$ and $\mathcal{A}_F \subset \mathcal{A}_{F'}$. We assume that the tests are closed under finite joint observations: after replacing $\mathscr{T}$ by its finite-joint closure, each Q_F is itself an allowed test (up to a Borel isomorphism of its target). This replacement does not change the generated sigma-field, and it ensures that convergence for all cylinder gauges controls the joint laws $(Q_F)_{\#}\mu$, not merely the separate one-test marginals. We also assume that the tests are generating in the sense that

$$\sigma\left(\bigcup_{F \Subset \mathscr{T}} \mathcal{A}_F\right) = \mathcal{A} \quad \text{modulo } \mu + \nu \tag{4.1}$$

for every pair of laws under consideration. A fixed increasing sequence of finite tests suffices when $\mathcal{A}$ is countably generated, but the directed formulation avoids choosing coordinates.

For probability laws a, b, let

$$\operatorname{Aff}(a, b) := \int \sqrt{\frac{da}{d\lambda}\frac{db}{d\lambda}}\, d\lambda, \qquad h(a, b)^2 := 2\{1 - \operatorname{Aff}(a, b)\}, \qquad \lambda := a + b.$$

The expression is independent of the dominating law. There are two natural finite-test quantities, and they have different jobs.

Definition 4.1 (Cylinder gauge)**.** For $F \Subset \mathscr{T}$, define

$$d_F^{\mathrm{cyl}}(\mu, \nu) := \max_{\tau \in F} h(\tau_{\#}\mu, \tau_{\#}\nu). \tag{4.2}$$

The family $(d_F^{\mathrm{cyl}})_F$ generates the initial, or cylinder, uniformity on probability laws.

The gauges are monotone under inclusion:

$$F \subset F' \quad \Longrightarrow \quad d_F^{\mathrm{cyl}} \le d_{F'}^{\mathrm{cyl}}.$$

Their role is topological. They say that finitely many prescribed marginals are close, and finite maxima allow different menus of tests to be combined.

Definition 4.2 (Joint projection energy)**.** For $F \Subset \mathscr{T}$, define

$$E_F(\mu, \nu) := -2 \log \operatorname{Aff}\big((Q_F)_{\#}\mu, (Q_F)_{\#}\nu\big), \tag{4.3}$$

with the convention $-\log 0 = +\infty$.

The energy is the order-1/2 Rényi divergence of the joint observation. It is not the maximum of the one-test divergences. The latter discards dependence among the tests and therefore cannot represent information in the new shell $\mathcal{A}_{F'} \ominus \mathcal{A}_F$. Nor do we assert that E_F is a metric: it is a monotone action used to exhaust a measure class. In short,

$$\boxed{d_F^{\mathrm{cyl}} \text{ generates the finite-test uniformity,} \qquad E_F \text{ measures joint information and its increments.}}$$

4.2 Projection limits and singularity explosion

Write $\mu_F = (Q_F)_{\#}\mu$, $\nu_F = (Q_F)_{\#}\nu$, and $\rho_F = \mathrm{Aff}(\mu_F, \nu_F)$. Data processing immediately gives $\rho_{F'} \le \rho_F$, and hence $E_{F'} \ge E_F$, whenever $F \subset F'$. The limit has an exact global meaning.

Theorem 4.3 (Projection affinity theorem)**.** *Under* (4.1)*, as a directed net,*

$$\rho_F \downarrow \mathrm{Aff}(\mu, \nu), \qquad E_F(\mu, \nu) \uparrow -2 \log \mathrm{Aff}(\mu, \nu). \tag{4.4}$$

Consequently,

$$\mu \perp \nu \quad \Longleftrightarrow \quad E_F(\mu, \nu) \longrightarrow +\infty. \tag{4.5}$$

Proof. Set $\lambda = (\mu + \nu)/2$, $p = d\mu/d\lambda$, and $q = d\nu/d\lambda$. Pulling finite-stage densities back to X gives

$$p_F = \mathbb{E}_\lambda[p \mid \mathcal{A}_F], \qquad q_F = \mathbb{E}_\lambda[q \mid \mathcal{A}_F].$$

The generating assumption and martingale approximation imply $p_F \to p$ and $q_F \to q$ in $L^1(\lambda)$. Since

$$\|\sqrt{a} - \sqrt{b}\|_{L^2}^2 \le \|a - b\|_{L^1},$$

we obtain $\int \sqrt{p_F q_F}\, d\lambda \to \int \sqrt{pq}\, d\lambda$. Monotonicity follows from data processing. Finally, affinity is zero exactly when the two laws are mutually singular. $\square$

There is also an exact conditional increment. If $F \subset F'$ and $\rho_F > 0$, disintegrate the two F'-stage laws over the common F-stage observation. Let $r_{F'|F}(x)$ be the conditional affinity and let η_F be the normalized Hellinger midpoint law, defined by

$$\frac{d\eta_F}{d(\mu_F + \nu_F)} = \frac{\sqrt{d\mu_F/d(\mu_F + \nu_F)}\sqrt{d\nu_F/d(\mu_F + \nu_F)}}{\rho_F}.$$

Then

$$\rho_{F'} = \rho_F \int r_{F'|F}\, d\eta_F, \qquad E_{F'} - E_F = -2\log \int r_{F'|F}\, d\eta_F \geq 0. \tag{4.6}$$

For independent product blocks the conditional affinity is independent of the old observation, and the energy increment becomes literally additive. In a general orbit, (4.6) is the replacement for an orthogonal sum of coordinate energies.

The Rényi–1/2 energy is preferable to relative entropy as a global class detector. Infinite relative entropy may occur for equivalent measures because a log density is not integrable. By contrast, (4.5) detects singularity with no integrability hypothesis.

4.3 Equivalence requires two-sided likelihood closure

Finite energy says that two laws are not mutually singular; without a dichotomy it does not say that they are equivalent. The missing condition is the preservation of likelihood mass in both directions.

Assume that $\mu_F \sim \nu_F$ at every finite stage and lift

$$Z_F := \frac{d\mu_F}{d\nu_F} \circ Q_F$$

to X. If $F \subset F'$, then

$$\mathbb{E}_\nu[Z_{F'} \mid \mathcal{A}_F] = Z_F. \tag{4.7}$$

Thus $(Z_F)_F$ is a nonnegative ν-martingale of mean one, while $(Z_F^{-1})_F$ is the corresponding μ-martingale.

Theorem 4.4 (Two-sided likelihood criterion)**.** *Under* (4.1) *and finite-stage equivalence,*

$$\mu \ll \nu \quad \iff \quad \{Z_F : F \Subset \mathscr{T}\} \quad \textit{is uniformly integrable in } L^1(\nu), \tag{4.8}$$

$$\nu \ll \mu \quad \iff \quad \{Z_F^{-1} : F \Subset \mathscr{T}\} \quad \textit{is uniformly integrable in } L^1(\mu). \tag{4.9}$$

In particular, $\mu \sim \nu$ if and only if both conditions hold.

Proof. If $Z = d\mu/d\nu$ exists in $L^1(\nu)$, then $Z_F = \mathbb{E}_\nu[Z \mid \mathcal{A}_F]$; conditional expectations of one L^1 variable form a uniformly integrable family. Conversely, uniform integrability makes the likelihood martingale relatively weakly compact in $L^1(\nu)$. Select a weakly convergent cofinal subnet $Z_{F_\alpha} \rightharpoonup Z$ in $L^1(\nu)$.

Cofinality means that for each fixed F_0 the indices eventually satisfy $F_\alpha \supset F_0$. Consequently, if $A \in \mathcal{A}_{F_0}$, then eventually

$$\int_A Z_{F_\alpha}\, d\nu = \int_A Z_{F_0}\, d\nu = \mu(A)$$

by (4.7); weak convergence against $\mathbf{1}_A \in L^\infty(\nu)$ gives $\int_A Z\, d\nu = \mu(A)$. The directedness of the finite test sets implies that $\bigcup_F \mathcal{A}_F$ is an algebra: if $A \in \mathcal{A}_F$ and $B \in \mathcal{A}_{F'}$, then both sets belong to $\mathcal{A}_{F\cup F'}$, which is closed under complements and finite unions. By (4.1), this algebra generates $\mathcal{A}$ modulo $\mu+\nu$. The monotone-class theorem therefore extends the identity to every $A \in \mathcal{A}$, so $d\mu = Z\, d\nu$. Interchanging the laws proves the reverse assertion. □

The reverse condition is genuinely a lower-tail condition for the forward likelihood:

$$\mathbb{E}_\mu\Big[Z_F^{-1}\mathbf{1}_{\{Z_F^{-1}>M\}}\Big] = \nu\{Z_F < M^{-1}\}. \tag{4.10}$$

It cannot be recovered from an upper-tail estimate on Z_F. A convenient quantitative version uses convex superlinear functions Φ, Ψ:

$$\sup_F \mathbb{E}_\nu \Phi(Z_F) < \infty, \qquad \sup_F \mathbb{E}_\mu \Psi(Z_F^{-1}) < \infty. \tag{4.11}$$

Together with compatible finite-stage limits and an honest realization, fixed de la Vallée–Poussin bounds give the two-sided uniform integrability needed to close both Radon–Nikodym directions. They do not by themselves supply the countably additive realization.

Lemma 4.5 (Hellinger lower semicontinuity of the Orlicz bounds)**.** *Let ν be a probability law, let $\lambda_i = z_i\nu$ be a net of probability laws with $z_i > 0$ ν-almost everywhere, and suppose that $\lambda_i \to \lambda$ in Hellinger distance. Then $\lambda = z\nu$ for some $z \geq 0$,*

$$\sqrt{z_i} \longrightarrow \sqrt{z} \quad \text{in } L^2(\nu), \qquad z_i \longrightarrow z \quad \text{in } L^1(\nu).$$

Let $\Phi, \Psi : [0,\infty) \to [0,\infty)$ be lower-semicontinuous convex functions, both superlinear at infinity. Define the lower-semicontinuous perspective

$$\Psi^{\leftarrow}(r) := \begin{cases} r\Psi(r^{-1}), & r > 0, \\ +\infty, & r = 0. \end{cases} \qquad \Psi^{\leftarrow}(0) = \lim_{r\downarrow 0} r\Psi(r^{-1}) = \lim_{u\to\infty} \frac{\Psi(u)}{u} = +\infty. \tag{4.12}$$

Then

$$\int \Phi(z)\, d\nu \le \liminf_i \int \Phi(z_i)\, d\nu, \tag{4.13}$$

$$\int \Psi^{\leftarrow}(z)\, d\nu \le \liminf_i \int \Psi^{\leftarrow}(z_i)\, d\nu = \liminf_i \int \Psi(z_i^{-1})\, d\lambda_i. \tag{4.14}$$

In particular, a finite uniform bound on the second functional forces $z > 0$ ν-almost everywhere and passes the reverse Orlicz bound to the limit.

Proof. Hellinger convergence implies convergence in total variation. Since every λ_i is absolutely continuous with respect to ν, so is λ. The Hellinger identity gives the asserted L^2 convergence of the square roots, and Cauchy–Schwarz then gives the L^1 convergence of the densities. The function $\Psi^{\leftarrow}$ is convex by the perspective construction and is lower-semicontinuous at zero by superlinearity. The integral of a nonnegative lower-semicontinuous convex function is lower-semicontinuous for L^1 convergence. This proves both inequalities. The equality in (4.14) is change of measure, and the value $\Psi^{\leftarrow}(0) = +\infty$ proves the final assertion. □

4.4 Why the bare projective uniformity is too weak

Let $\ell(dx) = \frac{1}{2}e^{-|x|}\, dx$ and $\mu_0 = \bigotimes_{j\ge1} \ell$ on $\mathbb{R}^{\mathbb{N}}$. Fix $c > 0$, $c \neq 1$, and define

$$T_n = \operatorname{diag}(\underbrace{c, \dots, c}_{n}, 1, 1, \dots).$$

Put $\mu_n = (T_n)_{\#}\mu_0$ and $\mu_\infty = (cI)_{\#}\mu_0$. Every μ_n is equivalent to μ_0, because only finitely many coordinates have changed. For each fixed finite coordinate set F, however,

$$(Q_F)_{\#}\mu_n = (Q_F)_{\#}\mu_\infty \quad \text{for all sufficiently large } n.$$

Thus (μ_n) is Cauchy, and in fact convergent to μ_∞, for the bare cylinder uniformity. The one-coordinate affinity $\rho = 2\sqrt{c}/(1+c)$ is strictly less than one, so

$$E_{\{1,\dots,n\}}(\mu_n, \mu_0) = -2n \log \rho = 2n \log \frac{1+c}{2\sqrt{c}} \longrightarrow \infty. \tag{4.15}$$

Kakutani's theorem gives $\mu_\infty \perp \mu_0$. Moreover, if $m > n$, then $\operatorname{Aff}(\mu_m, \mu_n) = \rho^{m-n}$; in particular, consecutive terms remain a fixed positive Hellinger distance apart. Hence the sequence is not Cauchy for the admissible metric defined below. There is no conflict: the completion of the bare cylinder uniformity contains the singular law μ_∞, whereas the stronger Cauchy structure below deliberately puts that boundary at nonzero distance.

4.5 The admissible metric completion

The example shows that selecting some cylinder-Cauchy threads after the fact would not produce a completion of the cylinder uniformity. We instead define a stronger, class-preserving Cauchy structure before taking a completion. Let

$$\mathcal{P}_{\mu_0} := \{\mu : \mu \text{ is a probability law and } \mu \sim \mu_0\}.$$

For $\mu \in \mathcal{P}_{\mu_0}$, write

$$Z_\mu := \frac{d\mu}{d\mu_0}, \qquad L_\mu := \log Z_\mu,$$

as equivalence classes modulo μ_0-null sets. On $L^0(\mu_0)$ use the complete metric

$$\delta_0(f, g) := \int_X \min\{1, |f - g|\}\, d\mu_0,$$

which induces convergence in μ_0-measure.

Definition 4.6 (Admissible Cauchy structure)**.** Define the likelihood-graph metric on $\mathcal{P}_{\mu_0}$ by

$$d_{\mathrm{LG},\mu_0}(\mu, \nu) := \delta_0(L_\mu, L_\nu) + \|Z_\mu - Z_\nu\|_{L^1(\mu_0)}. \tag{4.16}$$

If $H = \mathrm{Stab}_{\mathcal{G}}(\mu_0)$, the injective orbit map $\mathcal{G}/H \to \mathcal{P}_{\mu_0}$, $gH \mapsto g_\# \mu_0$, pulls (4.16) back to a Hausdorff metric and hence a specified uniformity on the orbit quotient. Fix a core $\mathcal{O}_{\mathrm{fin}} \subset \mathcal{P}_{\mu_0}$. A net in the core is *admissibly Cauchy* when it is Cauchy for d_{LG,μ_0}. Two admissibly Cauchy nets $(\mu_i)_i$ and $(\nu_k)_k$ are equivalent when

$$\lim_{(i,k)} d_{\mathrm{LG},\mu_0}(\mu_i, \nu_k) = 0 \tag{4.17}$$

on the product directed set. Denote the set of equivalence classes by $\widehat{\mathcal{O}}_{\mathrm{fin}}^{\mathrm{LG}}$ and put

$$\widehat{d}_{\mathrm{LG},\mu_0}\big([\mu_i], [\nu_k]\big) := \lim_{(i,k)} d_{\mathrm{LG},\mu_0}(\mu_i, \nu_k). \tag{4.18}$$

The double limit exists because both nets are Cauchy.

Theorem 4.7 (Completeness of the admissible Cauchy structure)**.** *The metric space $(\mathcal{P}_{\mu_0}, d_{\mathrm{LG},\mu_0})$ is complete. The formula (4.18) is therefore a well-defined complete metric, and the endpoint map identifies $\widehat{\mathcal{O}}_{\mathrm{fin}}^{\mathrm{LG}}$ isometrically with*

$$\overline{\mathcal{O}_{\mathrm{fin}}}^{\,d_{\mathrm{LG},\mu_0}} \subset \mathcal{P}_{\mu_0}. \tag{4.19}$$

Under this identification, the constant-net map is the inclusion of the core. In particular, the core is dense in its completion and every completion point *is an honest countably additive probability law equivalent to μ_0. Moreover, every admissibly Cauchy net is Cauchy for each cylinder gauge d_F^{cyl}.*

Proof. The map

$$\mathcal{J} : \mathcal{P}_{\mu_0} \longrightarrow L^0(\mu_0) \times L^1(\mu_0), \qquad \mathcal{J}(\mu) = (L_\mu, Z_\mu),$$

is an isometry when the product is given the sum of δ_0 and the L^1 metric. Its image is closed. Indeed, suppose $L_{\mu_n} \to L$ in measure and $Z_{\mu_n} \to Z$ in L^1. A common subsequence converges almost everywhere in both coordinates, and $Z_{\mu_n} = e^{L_{\mu_n}}$ yields $Z = e^L > 0$ almost everywhere. Also $\int Z \, d\mu_0 = 1$. Hence $\mu := Z\mu_0$ belongs to $\mathcal{P}_{\mu_0}$ and $\mathcal{J}(\mu) = (L, Z)$. Completeness of $L^0(\mu_0)$ for δ_0 and of $L^1(\mu_0)$ now proves completeness of $\mathcal{P}_{\mu_0}$.

The quotient of Cauchy nets by (4.17), equipped with the double-limit metric, is the standard metric completion. The endpoint map sends a class $[\mu_i]$ to the d_{LG,μ_0}-limit of (μ_i) in $\mathcal{P}_{\mu_0}$; it is a well-defined isometry onto the closed set (4.19). This proves the completion assertions.

Finally, data processing and $h(a,b)^2 \le \|da/d\lambda - db/d\lambda\|_{L^1(\lambda)}$ give

$$d_F^{\mathrm{cyl}}(\mu,\nu)^2 \le \|Z_\mu - Z_\nu\|_{L^1(\mu_0)} \le d_{\mathrm{LG},\mu_0}(\mu,\nu).$$

Thus admissible Cauchy nets are cylinder-Cauchy. Their finite-stage limits are compatible and are the projections of their unique global endpoint; by the generating assumption, equality of all those limits is equivalent to equality in the completed metric quotient. ☐

The Cauchy structure depends only on the reference measure class. Indeed, if $\nu_0 \sim \mu_0$, then changing reference subtracts the same fixed $\log(d\nu_0/d\mu_0)$ from every log-likelihood difference, the two L^1 terms agree by change of measure, and equivalent probability laws induce the same convergence-in-measure uniformity on L^0.

Projection energy and two-sided Orlicz bounds remain useful endpoint criteria, but they do not themselves define this completion. On an orbit with pairwise equivalence–singularity dichotomy, a uniform bound $\sup_F E_F(\mu,\mu_0) < \infty$ forces an already realized law μ into the class of μ_0. Without the dichotomy, the two-sided conditions (4.8)–(4.9) do the same, while lemma 4.5 passes fixed Orlicz bounds through finite-stage Hellinger limits. None of these criteria, without a Cauchy metric and a closed realization space, justifies calling a selected quotient of cylinder threads a completion.

4.6 An abstract projective Feldman–Hájek theorem

Theorem 4.8 (Conditional projective Feldman–Hájek principle)**.** *Suppose that every finite joint projection of every pair in $\mathcal{O}(\mu_0)$ is equivalent, the tests generate $\mathcal{A}$, and the orbit has the pairwise dichotomy*

$$\mu_g \sim \mu_k \quad \text{or} \quad \mu_g \perp \mu_k, \qquad \mu_g := g_{\#}\mu_0.$$

Then

$$\mu_g \sim \mu_k \quad \Longleftrightarrow \quad \sup_F E_F(\mu_g, \mu_k) < \infty, \qquad \mu_g \perp \mu_k \quad \Longleftrightarrow \quad E_F(\mu_g, \mu_k) \uparrow \infty. \tag{4.20}$$

If the dichotomy is absent, the first equivalence must be replaced by the two-sided UI conditions (4.8)–(4.9).

Proof. The projection affinity theorem identifies bounded energy with positive global affinity. Under the dichotomy, positive affinity rules out the singular branch and therefore forces equivalence. Divergence of the energy is exactly (4.5). The final statement is the two-sided likelihood criterion. □

The theorem is deliberately stated on the law side. It produces a finite energy component before any coordinates on $\mathcal{G}$ are chosen. An operator ideal can appear only after the energy has been calculated in a particular chart.

4.7 Hilbert information charts as a consequence

Suppose now that finite stages admit Hilbert coordinates. Let $(\mathsf{H}_F)_F$ be a directed system with isometric inclusions

$$\iota_{FF'} : \mathsf{H}_F \longrightarrow \mathsf{H}_{F'}, \qquad F \subset F',$$

and orthogonal bonding projections $\pi_{F'F} = \iota_{FF'}^* : \mathsf{H}_{F'} \to \mathsf{H}_F$, and let $\Gamma_F(g, k) \in \mathsf{H}_F$ be compatible pair coordinates. Assume

$$\pi_{F'F}\Gamma_{F'}(g, k) = \Gamma_F(g, k), \qquad F \subset F'.$$

Theorem 4.9 (Information-chart completion)**.** *Assume the hypotheses of the conditional projective theorem. If, on the orbit region under consideration, constants $0 < c \le C < \infty$ satisfy*

$$c\|\Gamma_F(g, k)\|^2_{\mathsf{H}_F} \le E_F(\mu_g, \mu_k) \le C\|\Gamma_F(g, k)\|^2_{\mathsf{H}_F} \tag{4.21}$$

uniformly in F, then

$$\mu_g \sim \mu_k \quad \Longleftrightarrow \quad \sup_F \|\Gamma_F(g,k)\|_{\mathsf{H}_F} < \infty. \tag{4.22}$$

Whenever these equivalent conditions hold, the compatible net converges in the Hilbert direct-limit completion H_{info}. Indeed, suppressing (g,k), compatibility and orthogonality give

$$\|\Gamma_{F'} - \iota_{FF'}\Gamma_F\|^2_{\mathsf{H}_{F'}} = \|\Gamma_{F'}\|^2_{\mathsf{H}_{F'}} - \|\Gamma_F\|^2_{\mathsf{H}_F}, \qquad F \subset F'. \tag{4.23}$$

Thus the squared norms form a bounded increasing net with limit $L := \sup_F \|\Gamma_F\|^2_{\mathsf{H}_F}$. If $F_0 \subset F$, $F_0 \subset G$, and $H \supset F \cup G$, then

$$\|\iota_{FH}\Gamma_F - \iota_{GH}\Gamma_G\|_{\mathsf{H}_H} \le \sqrt{\|\Gamma_H\|^2_{\mathsf{H}_H} - \|\Gamma_F\|^2_{\mathsf{H}_F}} + \sqrt{\|\Gamma_H\|^2_{\mathsf{H}_H} - \|\Gamma_G\|^2_{\mathsf{H}_G}} \le 2\sqrt{L - \|\Gamma_{F_0}\|^2_{\mathsf{H}_{F_0}}},$$

which tends to zero as F_0 increases. Hence the net is Cauchy and converges in H_{info}. If, more strongly,

$$c\|\Gamma_{F'} - \iota_{FF'}\Gamma_F\|^2_{\mathsf{H}_{F'}} \le E_{F'} - E_F \le C\|\Gamma_{F'} - \iota_{FF'}\Gamma_F\|^2_{\mathsf{H}_{F'}}, \qquad F \subset F', \tag{4.24}$$

then projection energy and Hilbert shell energy define the same Cauchy structure on the finite-stage core.

The Gaussian Hilbert–Schmidt space and the weighted ℓ^2 spaces of product models are instances of H_{info}; they are not inputs to the theorem. In a non-Gaussian matrix chart, the corresponding completion is determined by the Fisher operator computed in the preceding section.

One standard route to (4.24) is quadratic-mean differentiability. For a differentiable path $t \mapsto \mu_t$ with score $S_t \in L^2_0(\mu_t)$, the score visible to $\mathcal{A}_F$ is

$$S_{t,F} = \mathbb{E}_{\mu_t}[S_t \mid \mathcal{A}_F].$$

Orthogonal projection in $L^2(\mu_t)$ gives

$$\|S_{t,F'}\|_2^2 - \|S_{t,F}\|_2^2 = \|S_{t,F'} - S_{t,F}\|_2^2, \qquad F \subset F'. \tag{4.25}$$

This identifies the infinitesimal information shell. Uniform DQM remainder bounds are still required to pass from (4.25) to the finite Hellinger comparison (4.24); local Fisher calculus by itself does not close a global measure class.

4.8 Changing the finite test system

The cylinder uniformity is intrinsic provided two generating test systems compare cofinally. If p_F and q_G are their finite-stage gauges, assume that for every finite F and every $\varepsilon > 0$ there are a finite G and $\delta > 0$ such that

$$q_G(\mu, \nu) < \delta \quad \Longrightarrow \quad p_F(\mu, \nu) < \varepsilon,$$

and assume the converse implication with p and q interchanged. Then the two families generate the same projective uniformity. The admissible metric (4.16) depends only on the reference law and the global Radon–Nikodym graph, not on the generating test system. Thus, for the same core, both test systems have exactly the same admissibly Cauchy nets and the same completion. If energy or two-sided UI blocks are used to verify endpoint equivalence, cofinality of those blocks transfers that verification between the two systems as well.

This is the nonstatistical content of finite-experiment comparison. A finite net reduces a compact menu of tests to finitely many representatives; a pullback compares those representatives; and the triangle inequality returns to the full menu. Uniform integrability supplies the closed realization of the limiting likelihoods. No Markov-kernel quotient is needed, and the null ideal of the original sample space is retained.

5 The quotient geometry of visible perturbations

The projection construction has a geometric interpretation. A probability law does not remember the automorphism used to produce it; it remembers only the automorphism modulo the exact symmetries of the reference law. The Fisher form is the infinitesimal metric of this quotient, while the projection energy determines which finite-length directions actually close to equivalent laws. These statements separate a local symmetry calculation from the global measure-class problem.

5.1 The right quotient by the exact stabilizer

Let $\mathcal{G}$ act on $(X, \mathcal{A})$ by bimeasurable automorphisms, and fix a probability law μ. Its exact stabilizer is

$$H_\mu := \mathrm{Stab}_{\mathcal{G}}(\mu) := \{h \in \mathcal{G} : h_\# \mu = \mu\}. \tag{5.1}$$

The orbit map $\pi_\mu(g) = g_{\#}\mu$ satisfies $\pi_\mu(gh) = \pi_\mu(g)$ for $h \in H_\mu$. Conversely, $g_{1\#}\mu = g_{2\#}\mu$ implies $g_2^{-1}g_1 \in H_\mu$. Therefore

$$\mathcal{G}/H_\mu \longrightarrow \mathcal{O}(\mu), \qquad gH_\mu \longmapsto g_{\#}\mu, \tag{5.2}$$

is a bijection onto the orbit. It is a right quotient, not a left quotient: precomposition by a symmetry of the source law leaves the pushforward unchanged.

The exact stabilizer should not be confused with the measure-class stabilizer

$$\mathcal{G}_{\mathrm{qi}}(\mu) := \{g \in \mathcal{G} : g_{\#}\mu \sim \mu\}.$$

The former identifies parameters that produce the same law. The latter is the finite-information component to be characterized. A Feldman–Hájek theorem computes $\mathcal{G}_{\mathrm{qi}}(\mu)/H_\mu$ inside $\mathcal{G}/H_\mu$; it must not erase H_μ by demanding proximity to the identity before passing to the quotient.

5.2 The root-likelihood orbit

For $g \in \mathcal{G}_{\mathrm{qi}}(\mu)$ define

$$\mathcal{R}_\mu(gH_\mu) := \sqrt{\frac{d(g_{\#}\mu)}{d\mu}} \in L^2(\mu). \tag{5.3}$$

This map is well defined and takes values in the positive part of the unit sphere. It is injective because equality of the square-root densities is equality of laws, hence equality of right cosets. Moreover,

$$\|\mathcal{R}_\mu(gH_\mu) - \mathcal{R}_\mu(kH_\mu)\|^2_{L^2(\mu)} = 2\{1 - \mathrm{Aff}(g_{\#}\mu, k_{\#}\mu)\}. \tag{5.4}$$

Thus the Hellinger geometry is an ordinary Hilbert geometry after embedding the quasi-invariant quotient orbit by square-root likelihoods. The projection energies of the preceding section are finite-observation exhaustions of this orbit.

The image of (5.3) need not be closed in the unit sphere. A strong $L^2(\mu)$ limit r of nonnegative unit roots still defines the honest probability law $r^2\mu$, so countable additivity is not lost at this stage. It can nevertheless lose the reverse Radon–Nikodym direction when r vanishes on a set of positive μ-measure, and the resulting law need not be produced by an automorphism. Failure of countable-additive realization is a separate danger for a merely projective marginal thread, before it has been embedded in this L^2 space. Orbit closedness and two-sided likelihood uniform integrability are therefore global hypotheses, not consequences of the quotient notation.

5.3 Fisher information and the tangent stabilizer

Assume now that the quasi-invariant subgroup $\mathcal{G}_{\mathrm{qi}}(\mu)$ carries a Lie structure near H_μ, or that we have chosen a restricted Lie subgroup of it on which the following derivatives are defined, and assume that H_μ is a (split, in the infinite-dimensional case) Lie subgroup. Let $\mathfrak{g}_{\mathrm{qi}}$ denote the resulting Lie algebra. Suppose that the root-likelihood orbit map is differentiable in quadratic mean at the identity: for $A \in \mathfrak{g}_{\mathrm{qi}}$, the path $t \mapsto \exp(tA)_{\#}\mu$ has score $S_A \in L^2_0(\mu)$, and $A \mapsto S_A$ is the resulting linear differential. Define

$$\mathcal{I}_\mu(A, B) := \mathbb{E}_\mu[S_A S_B]. \tag{5.5}$$

Differentiating the root likelihood gives

$$D\mathcal{R}_\mu(eH_\mu)[A] = \frac{1}{2} S_A, \qquad \|D\mathcal{R}_\mu(eH_\mu)[A]\|_2^2 = \frac{1}{4}\mathcal{I}_\mu(A, A). \tag{5.6}$$

Proposition 5.1 (Fisher kernel and exact symmetries)**.** *Whenever the above DQM derivatives exist,*

$$\mathrm{Lie}(H_\mu) \subseteq \ker \mathcal{I}_\mu. \tag{5.7}$$

If the induced root-likelihood map $\mathcal{G}_{\mathrm{qi}}(\mu)/H_\mu \to L^2(\mu)$ is an immersion at the identity coset—the differential form of local identifiability—then equality holds:

$$\ker \mathcal{I}_\mu = \mathrm{Lie}(H_\mu). \tag{5.8}$$

In that case $\mathcal{I}_\mu$ descends to a positive definite quadratic form on $\mathfrak{g}_{\mathrm{qi}}/\mathrm{Lie}(H_\mu)$.

Proof. If $A \in \mathrm{Lie}(H_\mu)$, the path $\exp(tA)_{\#}\mu$ is constant, so its score is zero. This proves the inclusion. If the quotient orbit map is an immersion, a zero score means that the tangent class of A in $\mathfrak{g}_{\mathrm{qi}}/\mathrm{Lie}(H_\mu)$ is zero, proving the reverse inclusion. □

The inclusion is automatic; equality is not. A parametrization can be locally identifiable while having a singular differential, and a DQM score can vanish along a higher-order but nonconstant path. The immersion assumption rules out precisely these pathologies.

5.4 Gaussian symmetry after whitening

For the standard Gaussian law in $\mathbb{R}^n$, the exact linear stabilizer is $O(n)$ and the law orbit is

$$\mathrm{GL}(n)/O(n).$$

If $T_t = I + tA$, direct differentiation gives

$$S_A(x) = x^\top \operatorname{Sym}(A)x - \operatorname{tr} A, \qquad \mathcal{I}_\gamma(A, A) = 2\|\operatorname{Sym}(A)\|_F^2.$$

Hence

$$\ker \mathcal{I}_\gamma = \mathfrak{so}(n) = \operatorname{Lie}(O(n)), \qquad \mathfrak{gl}(n)/\mathfrak{so}(n) \simeq \operatorname{Sym}(n).$$

The skew directions are invisible because they are tangent to an exact symmetry orbit, not because their Hilbert–Schmidt norm vanishes.

For a positive injective Gaussian covariance C, the unwhitened invisible condition for a transformation tangent A is

$$AC + CA^* = 0. \tag{5.9}$$

In finite dimensions, whitening gives $B = C^{-1/2}AC^{1/2}$, and (5.9) becomes $B + B^* = 0$. The same statement holds in infinite dimensions for admissible tangents satisfying $A \operatorname{Ran}(C^{1/2}) \subseteq \operatorname{Ran}(C^{1/2})$ for which $C^{-1/2}AC^{1/2}$ extends to a bounded operator on the whitened Cameron–Martin source. Thus "skew" means skew in the whitened information metric; ordinary skew-adjointness in ambient coordinates is generally wrong.

In the restricted infinite-dimensional Gaussian group, the same quotient calculation leaves the symmetric Hilbert–Schmidt tangent directions (and the Cameron–Martin directions when translations are included). The global Gaussian equivalence theorem is stronger: determinant identities and the Gaussian dichotomy integrate this local quotient metric all the way to an endpoint criterion. That integration is additional structure, not a formal consequence of $O(H)$ being infinite dimensional.

5.5 Generic product laws recover the skew directions

Let $\mu_n = f^{\otimes n}$, where $f(x) = Z^{-1}e^{-V(x)}$ is even, centered, has variance one, and satisfies the integration-by-parts and moment assumptions of the preceding section. Put

$$s = V', \qquad J = \mathbb{E}[s(X)^2], \qquad \kappa = \mathbb{E}[(s(X)X - 1)^2].$$

For the path $I + tA$, decompose

$$A = D + S_0 + K, \quad D = \operatorname{diag} A, \quad S_0 = \frac{A + A^*}{2} - D, \quad K = \frac{A - A^*}{2}.$$

The Fisher form is represented by an operator diagonal on the orthogonal decomposition $\mathfrak{d}_n \oplus \mathrm{Sym}_{0,n} \oplus \mathfrak{so}(n)$:

$$\mathscr{I}_f = \kappa P_{\mathfrak{d}_n} + (J+1)P_{\mathrm{Sym}_{0,n}} + (J-1)P_{\mathfrak{so}(n)}. \tag{5.10}$$

Equivalently,

$$\mathcal{I}_f(A, A) = \kappa \|D\|_F^2 + (J+1)\|S_0\|_F^2 + (J-1)\|K\|_F^2.$$

The three eigenvalues have multiplicities n, $n(n-1)/2$, and $n(n-1)/2$, respectively.

The information inequality gives $J \geq 1$, with equality, under the stated regularity, only for the Gaussian density. Thus the Gaussian point is the spectral degeneracy $J - 1 = 0$. For a genuinely non-Gaussian density with $J > 1$ and $\kappa > 0$, the Fisher form sees the entire matrix tangent space and is dimension-free equivalent to the Frobenius norm:

$$\min\{\kappa, J-1\}\|A\|_F^2 \leq \mathcal{I}_f(A, A) \leq \max\{\kappa, J+1\}\|A\|_F^2. \tag{5.11}$$

This does not mean that the exact stabilizer is trivial. An i.i.d. product law is unchanged by coordinate permutations; if f is even, it is also unchanged by coordinatewise sign changes. In each fixed finite-dimensional block, the generic non-Gaussian linear stabilizer is a finite signed-permutation group and hence has zero Lie algebra, consistently with (5.11). The full infinite permutation group need not be discrete in every operator topology, but it supplies no smooth mixing direction in the restricted Lie chart used here. It remains essential globally: an infinite permutation can preserve the law exactly while $Q - I$ is not Hilbert–Schmidt. Any endpoint criterion written as $T - I \in \mathcal{S}_2$ before quotienting by H_μ is therefore false even in the i.i.d. setting.

5.6 An exact block-rotation model

The distinction between tangent visibility and global quotient geometry is already exact for independent two-dimensional blocks. Let $p = f \otimes f$ and let R_θ be planar rotation by angle θ. Define the rotation stabilizer and one-block energy by

$$H_{\mathrm{rot}} := \{\theta \in \mathbb{R}/2\pi\mathbb{Z} : (R_\theta)_\# p = p\}, \qquad e(\theta) := -2 \log \mathrm{Aff}((R_\theta)_\# p, p).$$

For a generic even non-Gaussian f, H_{rot} is the four-element subgroup generated by $\pi/2$; special densities may have a larger stabilizer. On $X = (\mathbb{R}^2)^{\mathbb{N}}$

consider

$$\mu = p^{\otimes \mathbb{N}}, \qquad T_\theta = \bigoplus_{j\geq 1} R_{\theta_j}.$$

Independence and the product dichotomy give the exact endpoint statement

$$(T_\theta)_\# \mu \sim \mu \quad \Longleftrightarrow \quad \sum_{j\geq 1} e(\theta_j) < \infty. \tag{5.12}$$

Near the identity coset, DQM and (5.10) yield

$$e(\theta) = \frac{J-1}{2}\theta^2 + o(\theta^2). \tag{5.13}$$

If the zeros of e are exactly H_{rot} and the orbit is proper away from those zeros, then (5.12) is equivalent to

$$\big(\mathrm{dist}(\theta_j, H_{\mathrm{rot}})\big)_{j\geq 1} \in \ell^2. \tag{5.14}$$

This is a complete non-Gaussian Feldman–Hájek criterion for the block model. It is an ℓ^2 condition on the quotient coordinates, not on the chosen representatives. For Gaussian f, every rotation belongs to the stabilizer, $e \equiv 0$, and all these directions disappear from the quotient.

5.7 What symmetry does and does not explain

Gaussian measure is therefore a high-symmetry, spectrally degenerate member of the general picture. Its infinite-dimensional orthogonal stabilizer removes every whitened skew direction, whereas a generic non-Gaussian product law has only signed-permutation symmetries in each finite linear block and assigns positive Fisher cost to infinitesimal rotations. This explains why the Gaussian coordinate expression uses only the symmetric Hilbert–Schmidt class and why the non-Gaussian tangent completion can involve the full Hilbert–Schmidt class.

Stabilizer dimension alone does not prove a global equivalence theorem. It determines the kernel of the local Fisher form under an immersion hypothesis, but it does not provide uniform nonlinear Hellinger estimates, projective realization, two-sided likelihood uniform integrability, orbit closedness, or an equivalence–singularity dichotomy. Those are exactly the additional ingredients isolated by the projection-increment construction. The robust statement is therefore hierarchical:

$$\boxed{\begin{array}{c} \text{exact stabilizer} \longrightarrow \text{quotient Fisher geometry} \\ \text{quotient Fisher geometry} \longrightarrow \text{projection-energy completion} \longrightarrow \text{global measure-class criterion} \end{array}}$$

The Gaussian theorem realizes every arrow. For the even i.i.d. density class in equation (3.1), theorem 3.1 realizes them for arbitrary bounded invertible linear mixing: dimension-free Hellinger coercivity and tail cumulant rigidity give necessity, while finite-dimensional entropy estimates and measurable Radon–Nikodym closure give sufficiency. The Fisher spectrum describes the local geometry; these separate arguments establish the endpoint.

References

[1] V. I. Bogachev. *Gaussian Measures*, volume 62 of *Mathematical Surveys and Monographs*. American Mathematical Society, 1998.

[2] R. H. Cameron and W. T. Martin. Transformations of wiener integrals under translations. *Annals of Mathematics*, 45(2):386–396, 1944.

[3] J. Feldman. Equivalence and perpendicularity of gaussian processes. *Pacific Journal of Mathematics*, 8(4):699–708, 1958.

[4] Jaroslav Hájek. On a property of normal distributions of any stochastic process. *Czechoslovak Mathematical Journal*, 8 (83)(4):610–618, 1958. In Russian.

[5] Svante Janson. *Gaussian Hilbert Spaces*, volume 129 of *Cambridge Tracts in Mathematics*. Cambridge University Press, Cambridge, 1997.

[6] Shizuo Kakutani. On equivalence of infinite product measures. *Annals of Mathematics*, 49(1):214–224, 1948.